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NOTES: Chapter 8: "Why Do the Monetary Authorities Smooth Interest

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Chapter 9: "Comment on Chapter 8" by Oimar Issing

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Call #: **HG925 .E963 1997**Location: **Shannon**

Book/Journal Title:

European Monetary Policy

Book Author:

Other Info:

Ch. 8 119-174
Ch. 9 175-178

Year: **1997**Pages: **119-178**

Article Author: **Collignon, Stefan;**
Association for the Monetary Union of Europe

Article Title: **Chapters 8 and 9**

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EUROPEAN MONETARY POLICY

Edited by
STEFAN COLLIGNON



PINTER

London and Washington

PINTER

A Cassell Imprint

Wellington House, 125 Strand, London WC2R 0BB

PO Box 605, Herndon, VA 20172

First published in 1997

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British Library Cataloguing in Publication Data

A catalogue record for this book is available from the British Library.

ISBN 1 85567 457 2

Library of Congress Cataloging-in-Publication Data

European monetary policy/edited by Stefan Collignon.

p. cm.

Papers presented at a conference of the Association for the Monetary Union of Europe, held in Kronberg, Nov. 30-Dec. 1 1995.

Includes bibliographical references and index.

ISBN 1-85567-457-2 (hc)

1. Monetary policy—European Union countries. 2. European currency unit. I. Collignon, Stefan, 1951- II. Association for the Monetary Union of Europe.

HG925.E963 1997

332.4'94-dc20

96-43601
CIP

ALD

HG

925

E963

1997

Typeset by BookEns Ltd., Royston, Herts.

Printed and bound in Great Britain by Biddles Ltd, Guildford and King's Lynn.

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CONTRIBUTORS

Fernando Barran is at the University of Louvain.

Stefan Collignon is Director of Research, Association for the Monetary Union of Europe, Paris.

Virginie Coudert is senior economist at the Centre d'Études Prospectives et d'Informations Internationales.

Alex Cukierman is in the School of Economics, Tel-Aviv University.

Spencer Dale is in the Department of Economics at the Bank of England, London.

Sylvester C. W. Eijffinger is Professor of Monetary Economics, Tilburg University and the College of Europe.

Peter M. Garber is at Brown University, Providence, Rhode Island.

Charles A.E. Goodhart is in the Financial Markets Group and the Department of Economics, London School of Economics and Political Science.

Andrew G. Haldane is in the Department of Economics, Bank of England, London.

Otmar Issing is a member of the Board of Directors, German Central Bank, Frankfurt.

Pierre Jaillet is Deputy Director, Monetary Research and Statistics Department, Bank of France, Paris.

Anil K. Kashyap is in the Graduate School of Business, University of Chicago.

Baron Alexandre Lamfalussy is President of the European Monetary Institute, Frankfurt.

Bennett T. McCallum is at Carnegie Mellon University, Pittsburgh.

Allan H. Meltzer is at the Carnegie Mellon University, Pittsburgh.

Franco Modigliani is Institute Professor Emeritus, Massachusetts Institute of Technology, Cambridge.

**Indicators of private non-financial sector balance sheet positions
(1993)**

	Net interest-bearing ^a assets of households (as a percentage of disposable income)	Household sector gross interest payments (as a percentage of income)	Net interest-bearing ^b assets of enterprises (as a percentage of GDP)	Enterprises' gross interest payments (as a percentage of income)
Belgium	168 ^f	3.9 ^d	-7 ^f	44 ^f
Denmark	-32 ^h	21.0 ^f	-34	57
Germany	48	5.0 ^e	-31 ^e	30 ^{e,f}
Spain	51	3.6	-24	31
France	52	5.3	-53	35
Italy	142	5.5	-38	34
Netherlands	38	4.9 ^f	-47 ^g	31 ^f
UK	-2	7.8 ^b	-35 ^c	18
Austria	n.a.	2.7	n.a.	n.a.
Finland	6	4.4	-56	88

^aNet deposits, credits, bonds and money market instrument; ^bincluding unincorporated business; ^cnon-financial corporate business; ^d1991; ^eWestern Germany; ^f1992; ^g1987; ^hexcluding assets in pension funds.

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8

**WHY DO THE MONETARY AUTHORITIES
SMOOTH INTEREST RATES?**

Charles A. E. Goodhart

8.1 INTRODUCTION

Most central banks now give primacy to the control of inflation, and use their ability to control short-term interest rates as their main instrument to that end. They have not, however, achieved this goal very effectively in past decades. As will be documented subsequently, inflation in most major countries has been only marginally mean reverting (stationary), i.e. the authorities have hardly been able to bring it back to some 'desired' equilibrium following an (inflationary) shock. Moreover, in most countries, there is considerable persistence in such shocks.

A key reason for such relative failure, I shall assert, is that the authorities do not vary interest rates sufficiently aggressively, or promptly, to hold inflation to a desired path. This is commonly described by the phrase 'too little, too late'. As is demonstrated in the Appendix, if the authorities were really to give absolute primacy to inflation control, in an economy with persistence and lags, interest rates ought to be varied in the light of the latest observed shocks so as to stabilize future expected inflation. This would mean that interest rates should show the same pattern of autocorrelation as the shocks themselves. If the latter were random, so should be the former.

Instead, the authorities have a penchant for 'smoothing' interest rates. Instead of adjusting interest rates by a large enough jump, whenever

* I am grateful to Dr Haizhou Huang for his help in writing the Appendix, which is joint work; to Dr Otmar Issing and the other members of the AUME 1995 Kronberg Conference for their comments and suggestions; to Harald Kinzler and Richard Payne for patient and careful research assistance; and to the ESRC for financial support.

inflation begins to deviate from its desired path, the authorities prefer to make relatively small changes. Especially given the long lags in the operation of monetary policy, and the momentum that can build up within the inflationary process in the meantime, this means that such initial changes in interest rates are regularly too small to reverse the deviation. The consequence is, therefore, usually a series of relatively small interest rate changes in the same direction. Put another way, the series of both interest rate changes and fluctuations in inflation (around the desired level) are strongly autocorrelated.

The adoption of specific, quantified inflation targets, especially if combined with instrument independence, may provide an incentive to the central banks in such countries to vary their interest rates more aggressively in order to maintain inflation at the desired level. I intend to explore whether there is any evidence of this happening in a subsequent paper; and I shall also try to model this syndrome more formally in a separate paper jointly with Dr Huang.

In some large part this study was stimulated by an earlier exercise by Glenn Rudebusch (1995), who studied, *inter alia*, the interest rate setting behaviour of the Fed. Rudebusch (1995, p. 264) 'Characterize[s] the Fed's interest rate targeting behavior [of the Fed funds rate] during the sample period with these three attributes:

- FR1 *Transitory Daily Deviations.* Large deviations of the spot [Fed funds] rate from the target rate are allowed on a daily basis. However, deviations from the target rate are transitory and are largely eliminated by the following day.
- FR2 *Short-Term Interest Rate Smoothing.* Targets are adjusted in limited amounts at a restrained, deliberate pace, and target changes are seldom immediately reversed. Thus, a typical policy action is implemented over the course of several weeks with graduate increases or decreases (but not both) in the target rate.
- FR3 *Medium-Term Target Persistence.* Abstracting from short-term interest rate smoothing considerations (FR2), the target rate is set at a level the Fed expects to maintain. That is, beyond a horizon of about a month, there are no planned movements to react to information already known.'

Rudebusch's main interest is to examine whether such behaviour can help to explain the various findings on the extent to which the yield curve¹ can be used successfully to predict future changes in interest rates.² He leaves it, however, in his conclusion (pp. 271–2), to 'future research ... to understand why the Fed conducts policy in this manner'.

However, he does offer some speculations.

The short-run smoothing of interest rates (FR2) evident in the gradual target adjustments of limited size in a single direction may be of much greater import. The Federal Reserve as well as the financial press appears to interpret the purpose of such smoothing to be the avoidance of 'undue stress' on financial markets.

Thus, the *Wall Street Journal* (Wessel, 1994) quotes Fed Chairman Alan Greenspan as arguing that the central bank implemented 'measured and deliberate' increases in short-term rates in early 1994 'so as not to unsettle financial markets'. Besides gradual adjustments, interest rate smoothing also discourages quick reversals of the direction of target changes. Such reversals are thought to 'whipsaw' the market and also contribute to disorder (see Footnote 14 [which states 'Note the account in Meulendyke (1990, p. 41): "During most of the 1970s, the [Fed] was reluctant to change the funds rate by large amounts at any one time. ... Part of that reluctance reflected a wish to avoid short-term reversals of the rate. Keeping each rate adjustment small minimized the risk of overdoing the rate changes and then having to reverse course."']).

A similar rationale of stabilizing or steadying markets could also be given for the persistence described in attribute FR3. However, Goodfriend (1991) elucidates a more subtle reason why the Fed might impart random-walk behavior to the Fed funds rate. He argues that output and prices do not respond to daily fluctuations in the funds rate but only to rates of at least three- or six-months maturity. Thus, for the Fed to attain its macroeconomic goals, it must be able to manipulate these longer-term rates. However, such rates are determined by market expectations of future funds rates; thus, by presenting the markets with a clear path for the future funds rate, the Fed can influence the longer-term rates. A constant funds rate is the path that likely communicates policy intentions most clearly and perhaps most credibly to markets. Thus, the pursuit of macroeconomic stabilization may impart a high degree of persistence to the funds rate.

One of the purposes of the empirical exercise in section 8.2 is to discover how far the behaviour of the monetary authorities in the other countries in our sample, Australia, Germany, Japan and the United Kingdom, was similar to that reported by Rudebusch for the USA.³ To anticipate the statistical details, the answer is that there were many similarities. The two countries which appear to smooth interest rates the most (in a sense to be defined in section 8.2) are Germany and the USA, whereas the UK and Australia do so rather less. Since such smoothing, 'too little, too late', is often taken to be a cause of worse inflation, it is interesting that the countries with a worse inflationary experience smoothed less than the countries with a better record. There is, of course, a possibility of reverse causation. Countries facing more extreme shocks, and a worse record, may be forced to react more aggressively.⁴

In the Appendix, we try to take on the more difficult question of why the authorities smooth interest rates, as they are shown to do in section 8.3. We demonstrate that, in a relatively simple model, they should not act in this fashion. What aspects of this model are wrong? Do the authorities fail to adjust aggressively enough (to stabilize inflation) because they dislike large interest rate changes for their own sake, or because they fear that such large changes might cause a (temporary) deviation of real output (from its equilibrium level)? Can one distinguish between these motives?

Given the lags in the operation of monetary policy, the authorities should adjust interest rates today in response to the *forecast* level of inflation several quarters hence. There is little evidence that our central banks have successfully done so. Instead, as again reported in section 8.2,

interest rates are mostly adjusted after it becomes clear that inflationary pressures have altered.

We surmise, in section 8.3, that this may be because, in a world of uncertain forecasts, the central bank cannot obtain the broad political support that may be necessary to allow interest rates to be changed, in particular raised, (without a political, public and social uproar), until evidence of *actual*, rather than predicted, shifts in inflationary pressure are in the public domain. The exchange of views between the Chancellor of the Exchequer and Governor of the Bank of England in the early summer of 1995 in the UK can be interpreted in this light.

As noted earlier, however, the German monetary authorities are the most prone to such smoothing; this is despite their superior characteristics of independence, credibility and overall performance. And the Australian and UK authorities do so least among our sample of countries, despite being subject to greater political subservience. One possible explanation is that a central bank with more credibility can afford to react less aggressively in varying *nominal* interest rates, since, with the public having more stable inflationary expectations, a smaller change in nominal rates may translate into a large change in real rates. We have not, in the time available for this exercise, been able to test this hypothesis.

Another suggested explanation, offered by Goodfriend (1991), is that the adjustment process works via long-term interest rates, not short-term rates. Since the smoothing process is sufficiently regular to be predictable, long rates jump when short-term interest rates change direction, and do not (need to) move much further in response to subsequent smoothed changes, since these had been rationally anticipated (see Miles, 1994). There is some evidence (Cook and Hahn (1989) for the USA, and Dale (1993) for the UK) that the pattern of changes in short and long rates is consistent with this story. Dr Issing, and others, at the conference accepted that this behavioural pattern had some validity in the German case. We did seek to test this hypothesis empirically for Germany; the results are given at the end of section 8.2.

8.2 INTEREST RATE SMOOTHING: THE EMPIRICAL EVIDENCE

8.2.1 The characteristics of interest rates

The operation of monetary policy is undertaken in a broadly similar manner in all of the countries in our sample: Australia, Germany, Japan, UK and USA. The authorities use their monopoly control of the high-powered monetary base to influence a short-term money market rate. This can be varied in relatively fine steps, e.g. of fewer than five basis points in Germany, but can in some countries, especially in the USA, be subject to noise and shocks from sources other than the authorities themselves. For the USA we have taken Rudebusch's (1995) data for changes in the Federal funds rate target, 1974–9 (pp. 254–5) and 1984–92 (pp. 256–7). The intervening period was omitted since it covered the period of a different

Table 8.1

	Series	Source	Data Period	Periodicity
Australia	Call money rate	Borio and Fritz (1995)	1984–94	Monthly average
Germany	Allotment rate for open market Transactions (OMT) in securities under repurchase agreements	Bundesbank Monthly Report	1980–88	End daily
			1988–95	Monthly average
Japan	Collateralized overnight call rate	Bank of Japan Economic Statistics Annual	1960–95	Monthly average
UK	Band 2 bank bill rate	Bank of England Quarterly Bulletin and Monetary Statistics	1981–95	End daily

regime, targeting unborrowed reserves. For the remaining countries in our sample, we used the money market data shown in Table 8.1.⁵

Besides their control of the (finer) instrument of a money market rate, the authorities in each country also maintain a more formal, higher visibility, discount rate, which is varied less frequently, but in larger steps, with more of an announcement effect. This will generally be adjusted when the money market rate, following a series of small steps, deviates too far from its normal relationship with the discount rate and/or when the authorities want to give an overt signal of a change in their policy position. The specific series that we have used in each case are shown in Table 8.2.

We follow Rudebusch in starting our data period in 1974, whenever possible. But for all other than his series for the US Fed funds rate, we can

Table 8.2

	Series	Source	Data period
Australia	Rediscount rate ^a	Australian Reserve Bank Bulletin	1974–93
Germany	Discount rate	Bundesbank Monthly Report	1974–95
Japan	Discount rate	Bank of Japan Economic Statistics Annual	1974–95
UK	Base rate	Bank of England Financial Statistics	1974–95
USA	Discount rate	Federal Reserve Statistical Release H15	1962–95

^a Australia's rediscount rate ceased in 1993.

take either the two periods which overlap with his, or the full unbroken data period. In most cases we have done both, so that we can both compare our results with his, and get the benefit of working with a longer unbroken data set.

The normal format for showing on a chart how interest rates have moved is to present this on a calendar time basis. One such chart for the German discount rate is given as Figure 8.1, and similar charts for our other nine series (five countries, two rates) are available on request. For our purposes, however, the charts are more revealing when presented on an events basis, i.e. whenever a change in rates occurs independently of the time gap between such events. These charts, Figures 8.2 to 8.11, show much more clearly how smooth the series for such changes has been, especially so in Germany, and least in the UK.

In each case an interest rate change in one direction, up or down, is much more likely to be followed by a change in the same direction than by a reversal. This is shown in Table 8.3, which records the ratio of continuations (+ followed by +, or - by -) to reversals (+ followed by -, or - by +).

We regard this ratio (subject to a qualification below) as the best available single summary statistic of the tendency to smooth interest rates. The qualification is that in one instance, the German money market rate (OMT), there was a period, between 1983 and 1987, when this rate vibrated gently around an otherwise mostly level trend (see Figure 8.3). In this case calculation of the ratio of continuations/reversals greatly understates the smoothness of the series. Indeed, when the same statistic

Table 8.3: Ratio of reversals to total changes

	Ratio of reversals to total changes
US FF target rate (1974-9, 1984-92)	1 : 9.05
US discount rate (1974-9, 1984-92)	1 : 7.40
UK base rate (1974-9, 1984-92)	1 : 3.88
German discount rate (1974-9, 1984-92)	1 : 6.90
Japanese discount rate ^a (1974-9, 1984-92)	1 : 6.25
Japanese overnight call rate ^a (1974-9, 1984-92)	1 : 2.61
Australian rediscount rate (1974-9, 1984-92)	1 : 10.84
US discount rate (1962-95)	1 : 5.00
UK base rate (1974-95)	1 : 4.26
UK band 2 bank bill rates (1981-95)	1 : 3.91
German discount rate (1974-95)	1 : 7.33
German OMT rates ^a (1980-8)	1 : 2.09
German OMT rates ^a (1988-95)	1 : 7.67
Japanese discount rate ^a (1974-95)	1 : 9.00
Japanese overnight call rate ^a (1960-95)	1 : 2.88
Australian rediscount rate (1974-93)	1 : 6.93
Australian call money rate ^a (1984-94)	1 : 3.17

^a Only monthly averages available.

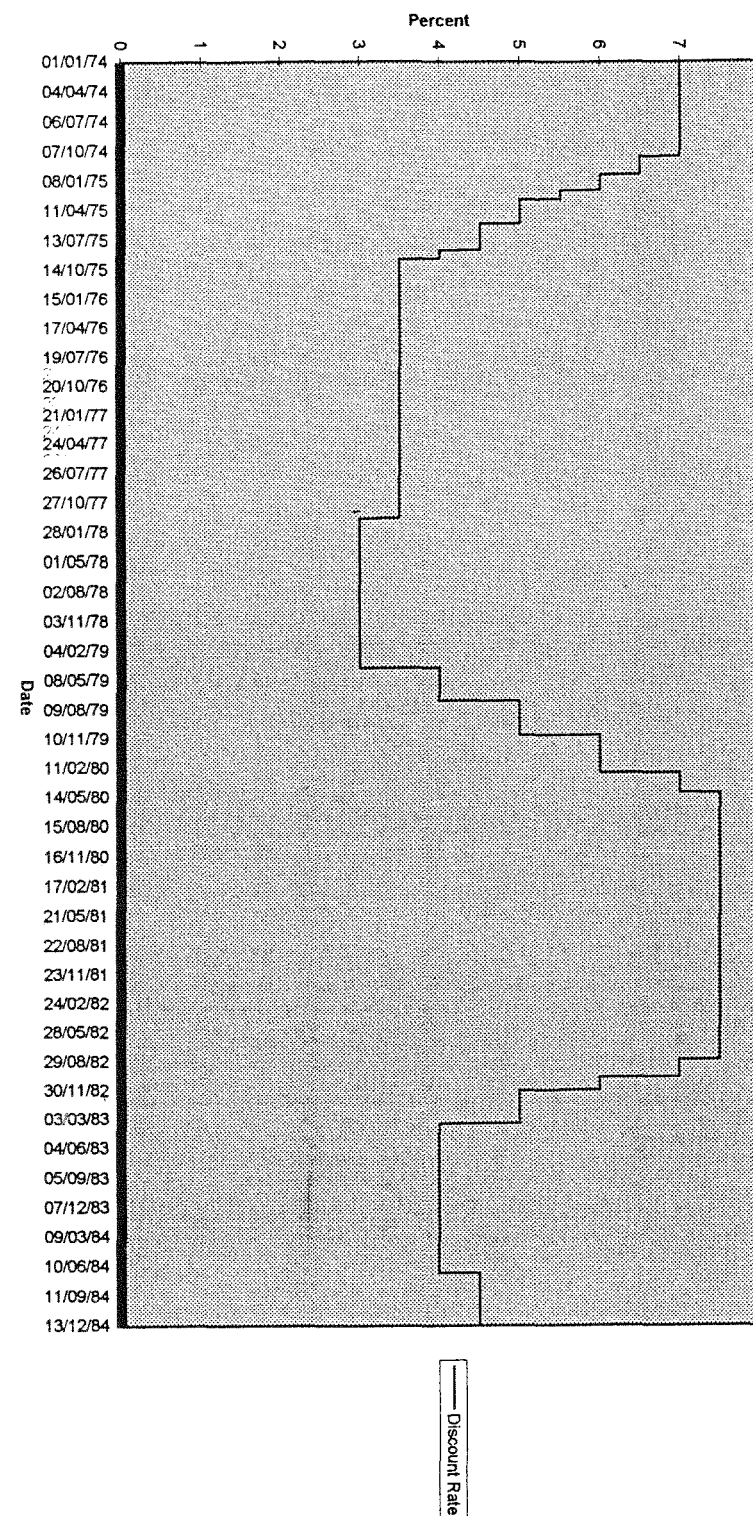
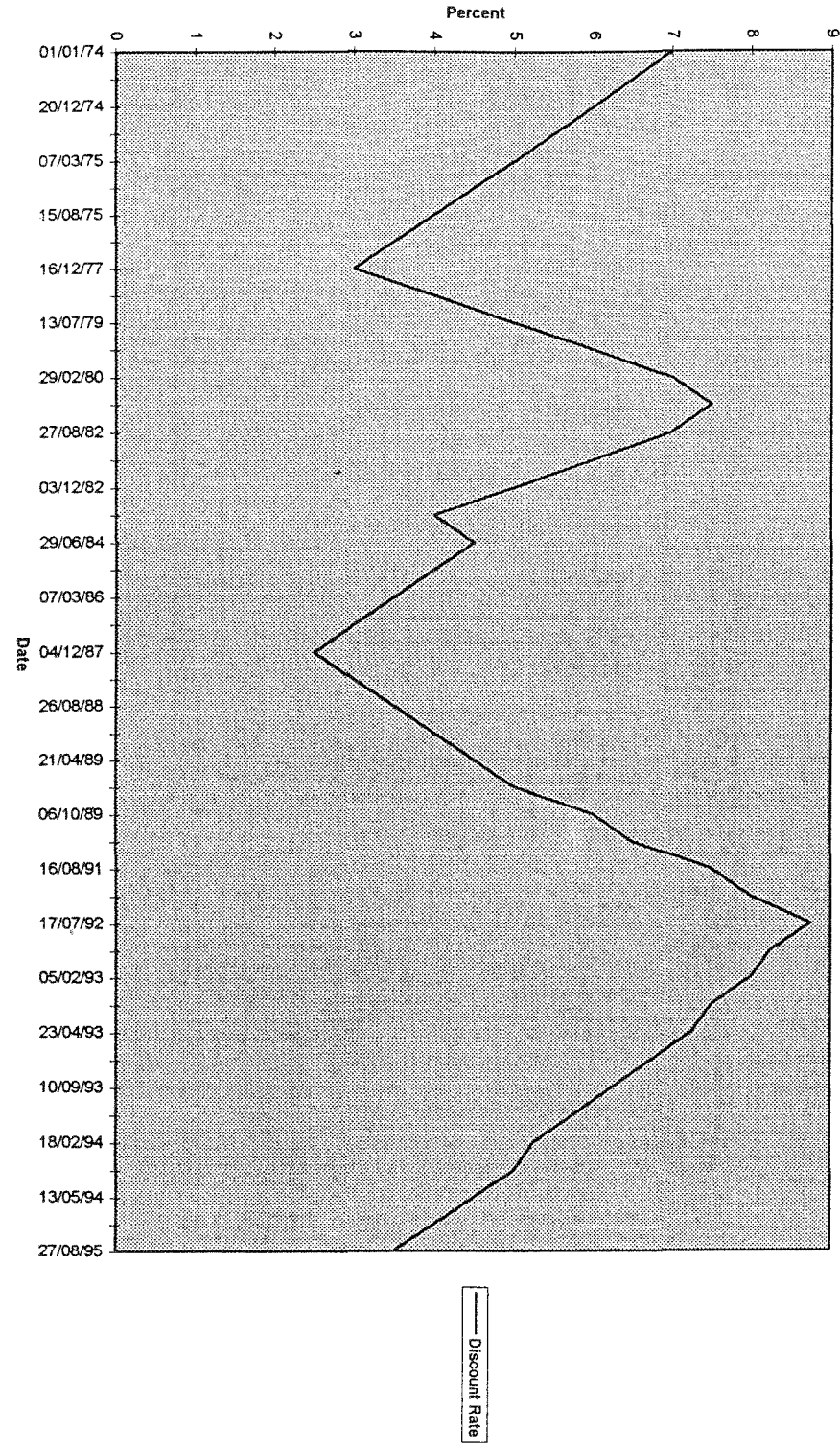
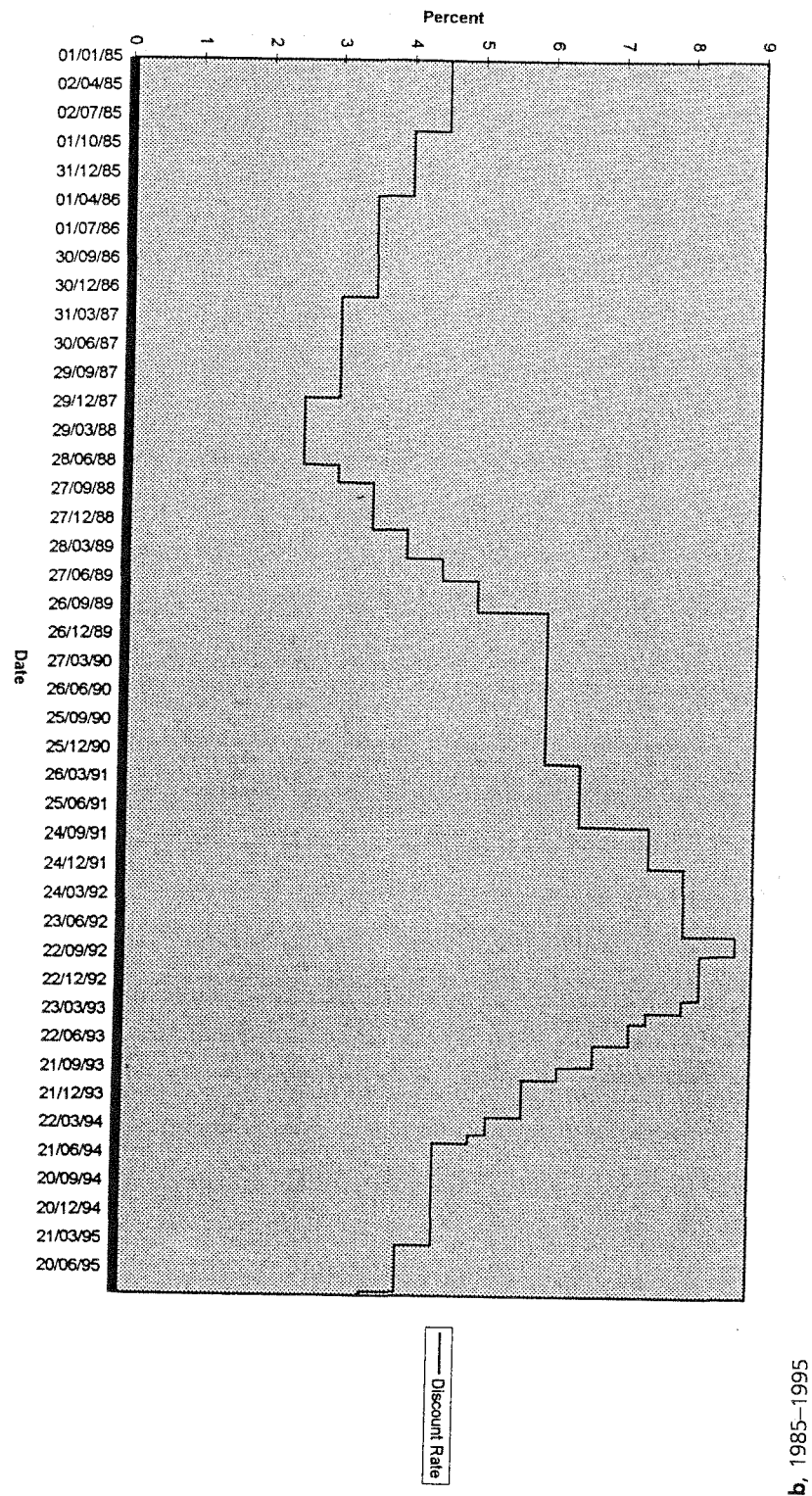


Figure 8.1 German daily discount rates, calendar based: a, 1974-1984;
Source: Deutsche Bundesbank Monthly Report.



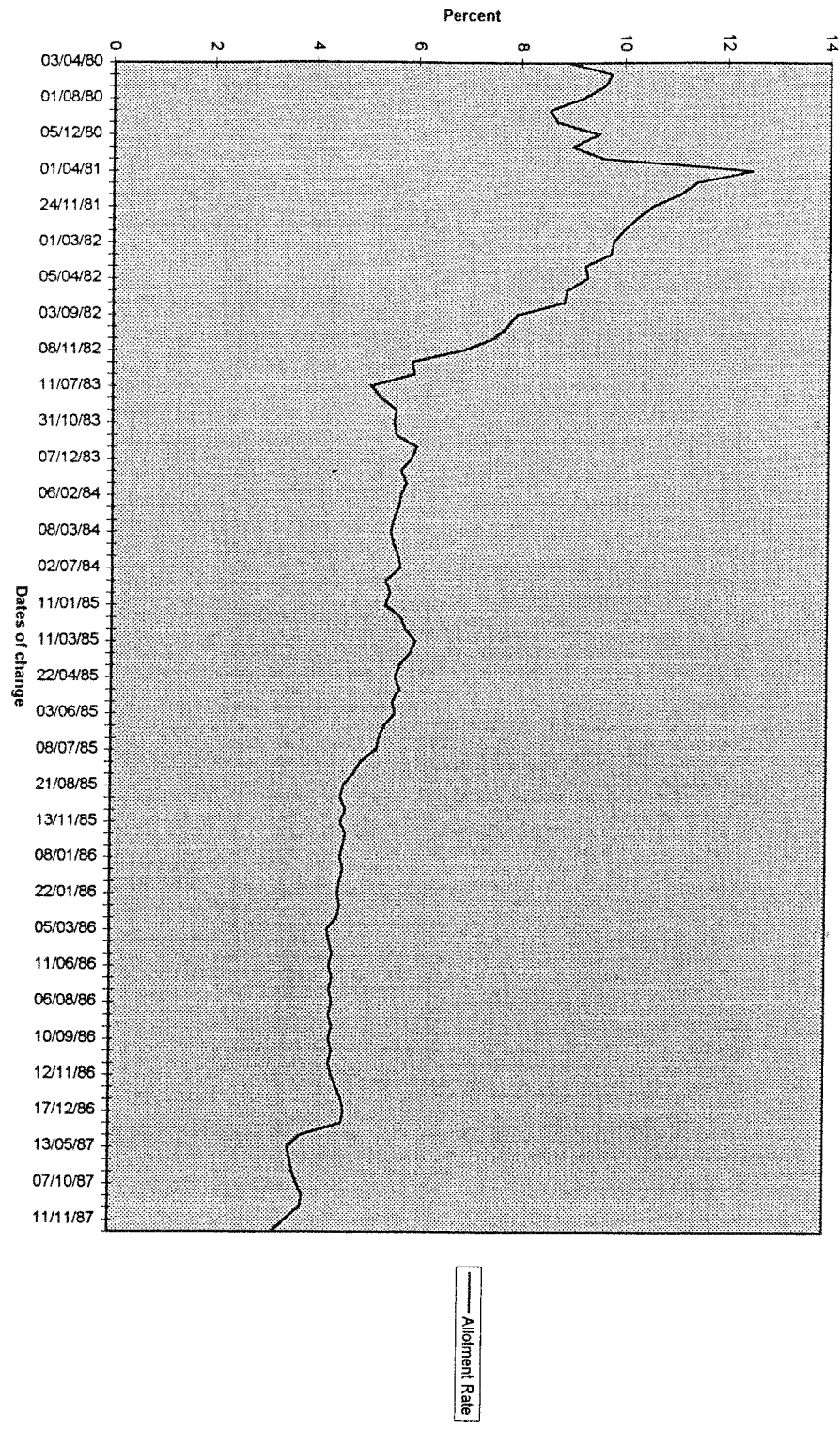
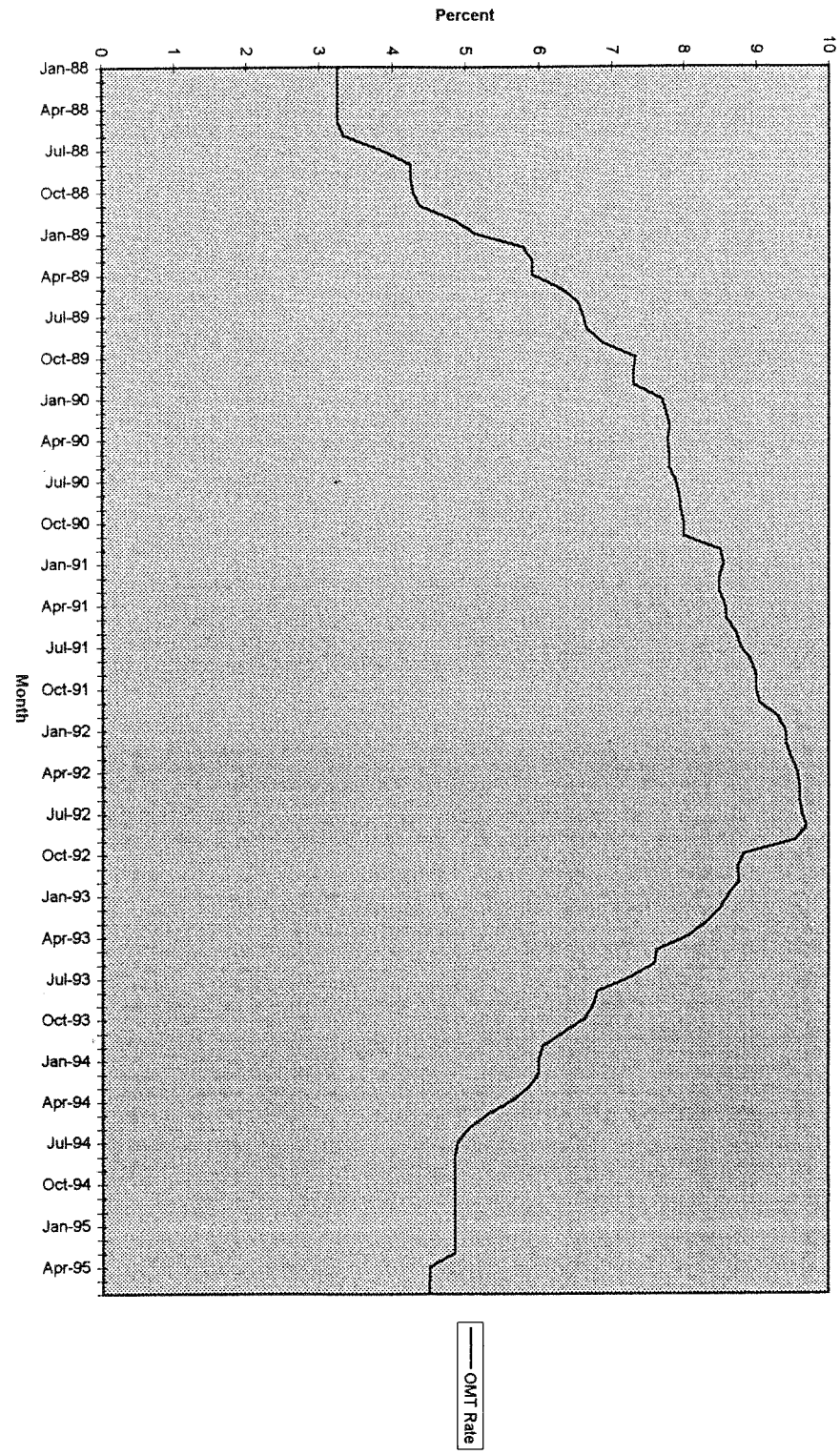


Figure 8.3 German allotment rate for open market transactions in securities under repurchase agreements: **a**, 1980–1987;
Source: Deutsche Bundesbank Monthly Report.



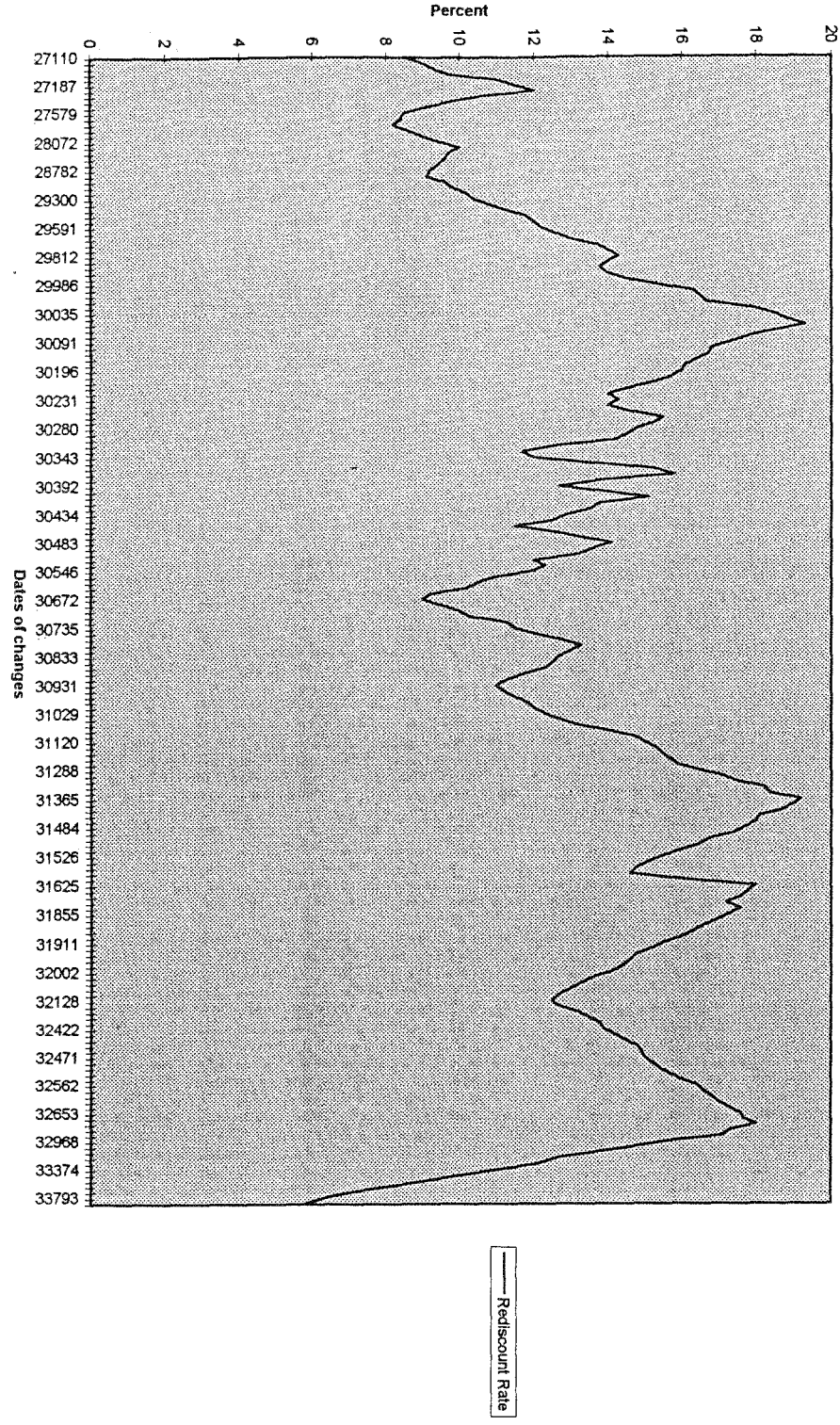


Figure 8.4 Australia rediscount rates, events based, 1974–1993
Source: Australian Reserve Bank Bulletin.

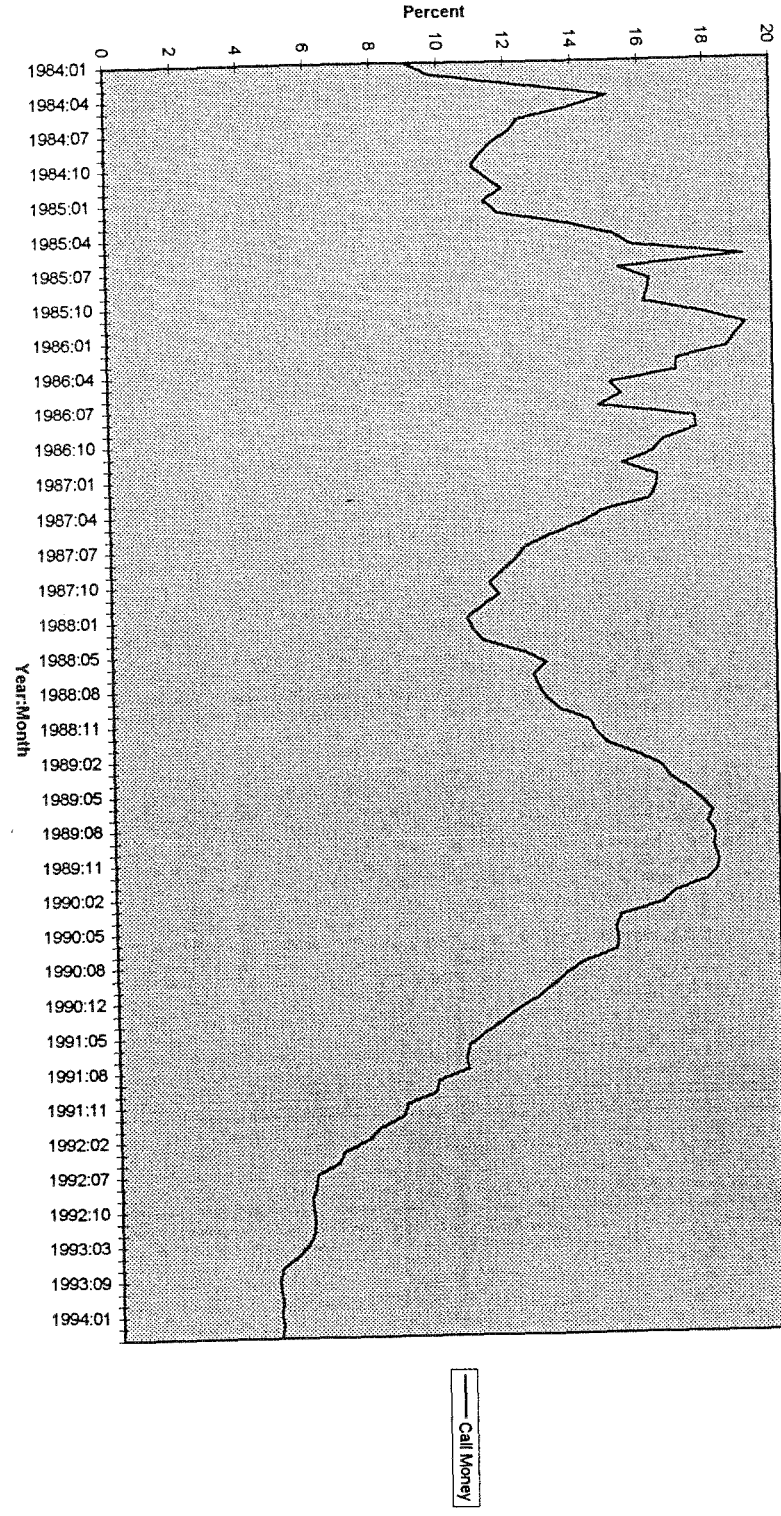


Figure 8.5 Australian call money rate, monthly averages, 1984–1994
Source: Borio and Fritz (1995).

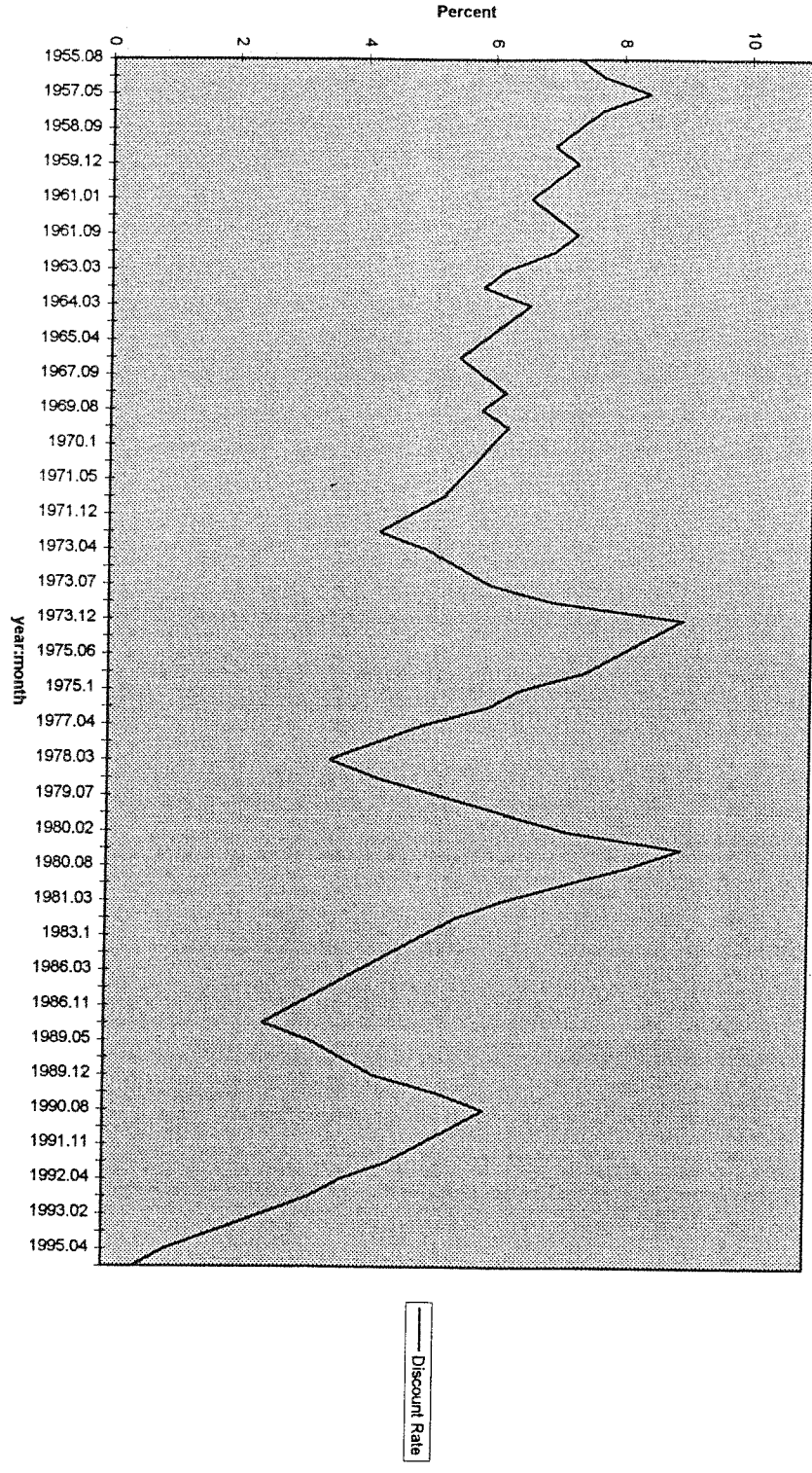


Figure 8.6 Japanese discount rate, end of month, 1955-1995
Source: *Economic Statistics Annual*, Bank of Japan.

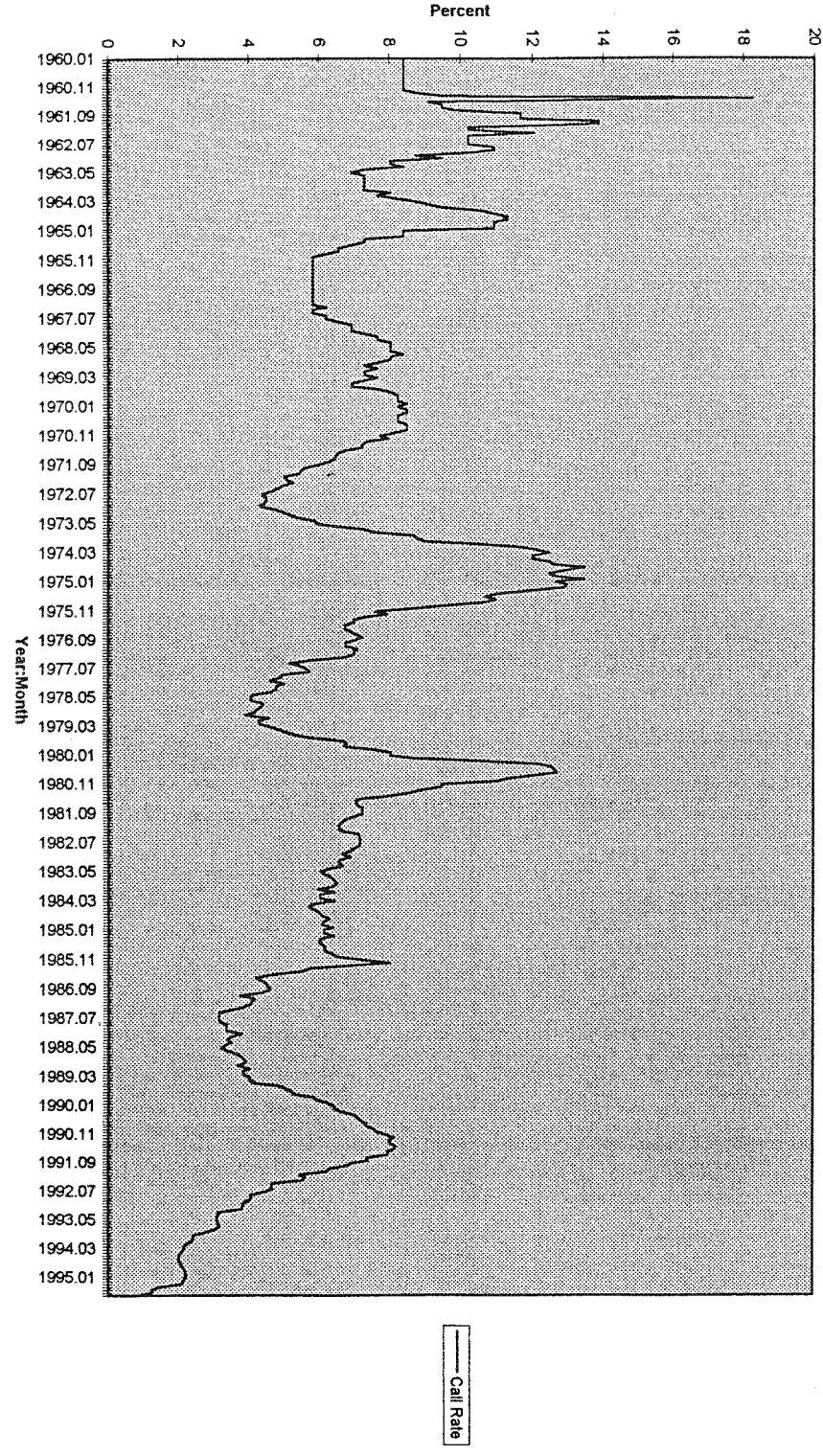


Figure 8.7 Japanese collateralized call rate, monthly average, 1960-1995
Source: *Economic Statistics Annual*, Bank of Japan.

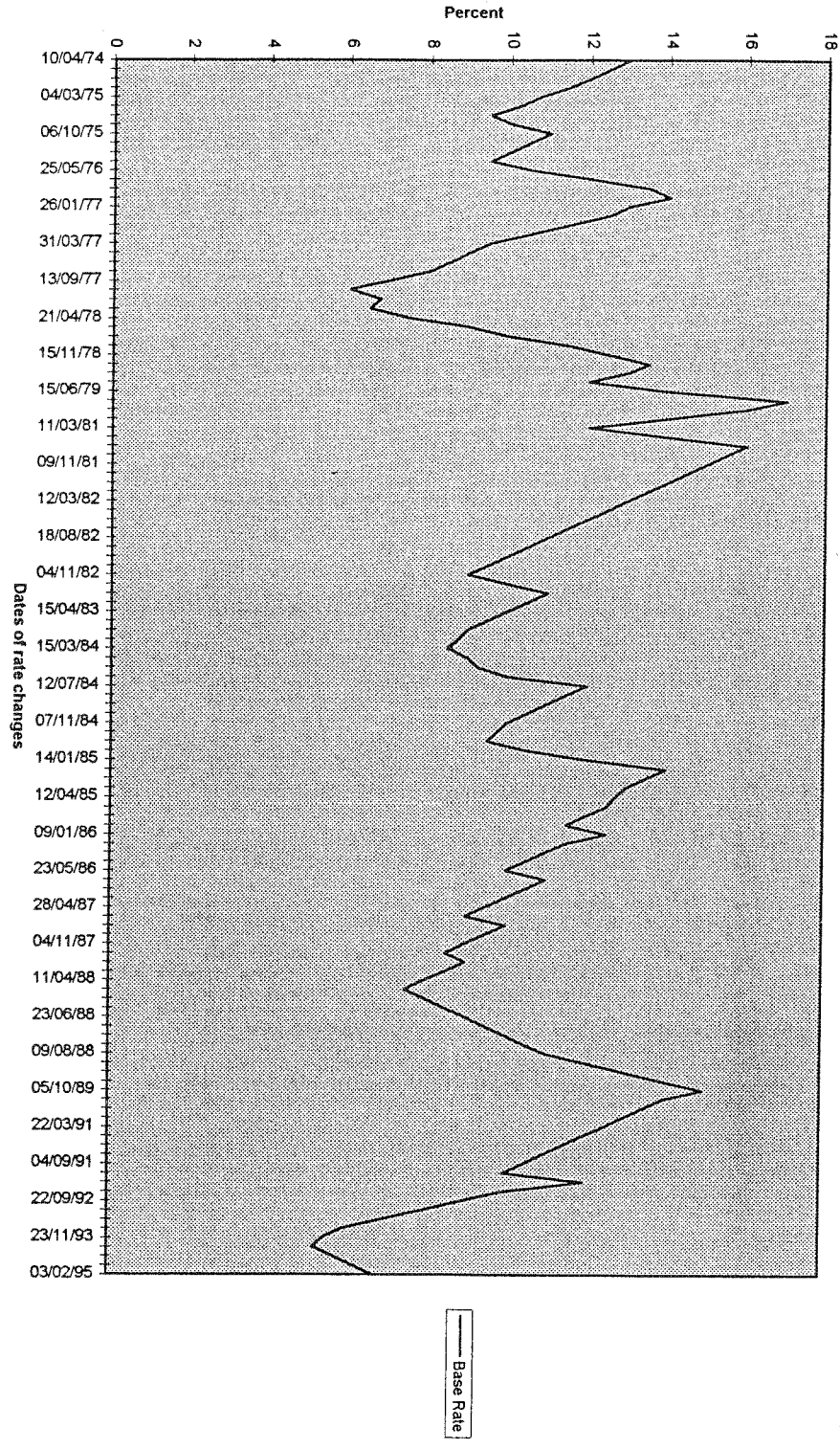


Figure 8.8 UK base rate, events based, 1974–1995
Source: Financial Statistics, Bank of England.

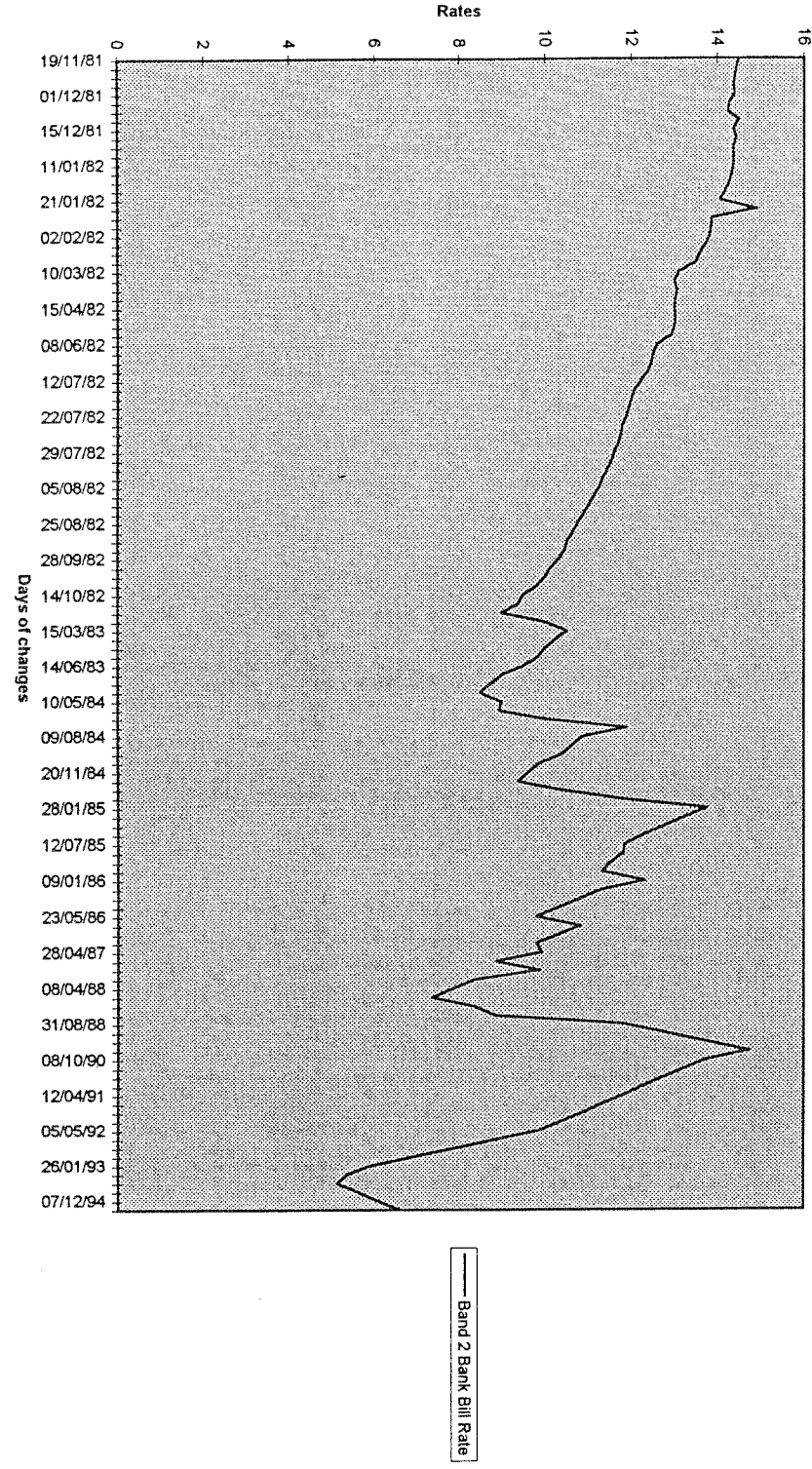


Figure 8.9 UK band 2 bill rates, events based, 1981–1995
Source: Bank of England Quarterly Bulletin.

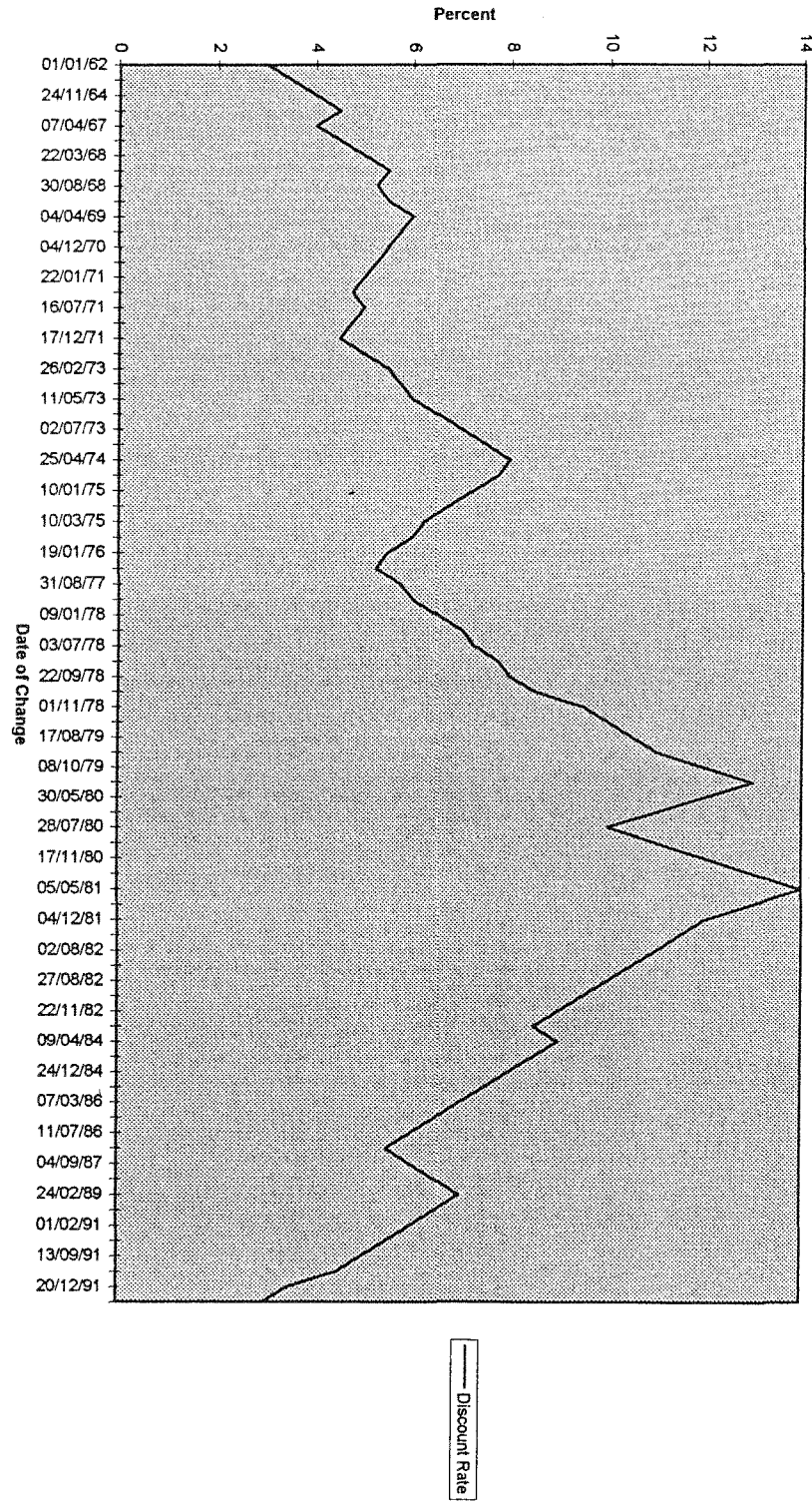


Figure 8.10 US discount rate, events based, 1962–1992
Source: Federal Reserve Statistical Release H15.

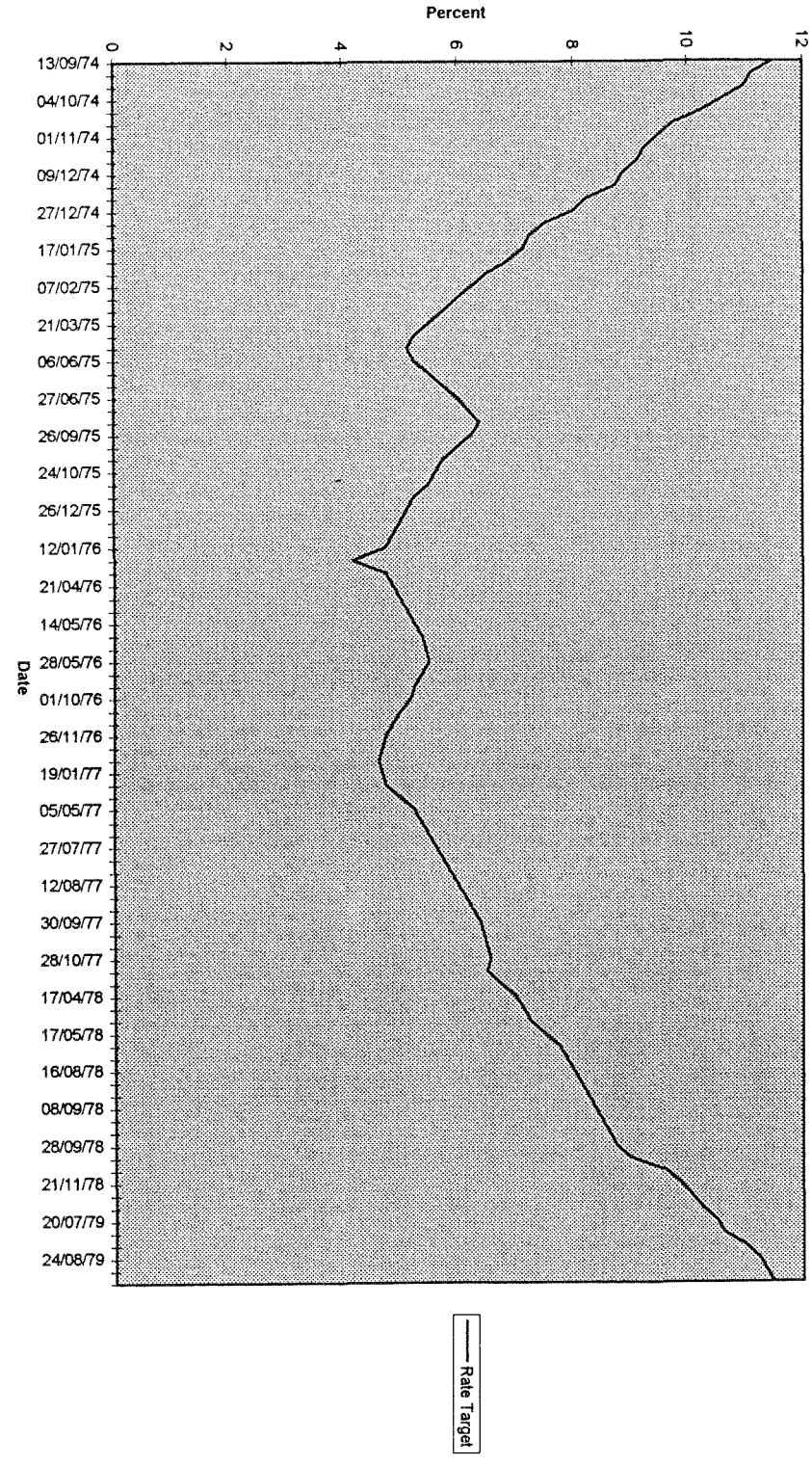
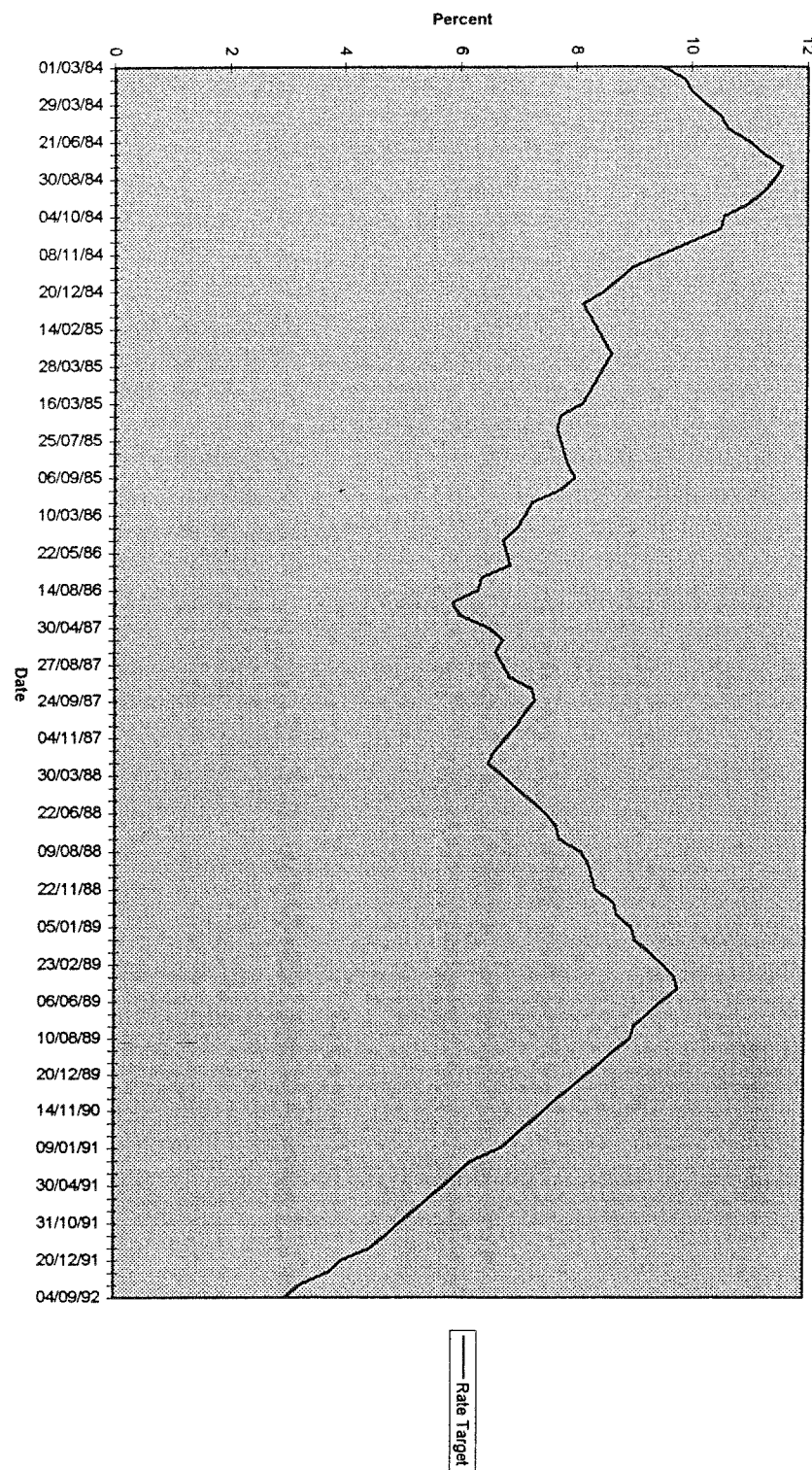


Figure 8.11 Rudebusch's original data of US Federal Funds rate target changes: a, 1974–1979;



b, 1984-1992

is calculated for the period 1988-95, when the periodicity of the data becomes monthly average, the German money market rate once again appears as the second smoothest by this metric. Also note that the US series in this case is Rudebusch's series for the Fed funds target rate, whereas the other series are actual data, and that some data are monthly average, while others are end-daily. Subject to these caveats, our ranking of relative smoothness for the money market series would be: the USA, Germany, Japan, the UK and Australia.

In Table 8.3 we first report the continuation/reversals ratio for the Rudebusch periods, and then for the full time period, wherever possible. In three cases, Australia, Japan and the US discount rate, the ratio is considerably lower for the full data period than in the Rudebusch periods, indicating that the years 1980-83 exhibited unusual volatility; but this was not apparently the case in the UK or Germany, where the full period ratio was marginally higher (smoother) than in the Rudebusch periods.

One reason often offered to explain smoothing in private sector financial markets, e.g. of earnings and dividends, is that the outcome for managers of the penalties inflicted by markets following reports of adverse results may outweigh the rewards from reports of good results (see Fudenberg and Tirole, 1995), e.g. owing to managers being risk averse. In such cases one might expect managers to be more reluctant to report bad results and/or cut dividends, so that, when the actual results could be massaged, there would be an asymmetry between the revelation of bad and good news. Bad news, e.g. dividend cuts, would be reported less frequently, and, when reported, would reveal a larger deviation from prior expectations than reports of good news.

Within the public sector, interest rate increases are treated as 'bad news'. By the same token one might expect interest rate increases to occur less regularly, but in larger jumps, than interest rate decreases. Moreover, in the UK it was conventional wisdom in the Bank of England that funding in the gilts market could not proceed successfully as long as the market feared further base rate increases, whereas, *per contra*, expectations of further cuts in interest rates would lead to a 'feeding frenzy' in the gilts market. This, again, could be expected to lead to an asymmetry whereby interest rate increases would be both fewer and larger than interest rate decreases.

In contrast, however, Rudebusch finds no evidence of any asymmetries, or significant differences, between his series of continuations (i.e. whether they were + followed by +, or - by -) or of reversals (+ - or - +). We examine this by looking in Table 8.4 at the average durations between changes,⁶ and the average size of changes, for each interest rate for each country/rate. Column 1 in each case reports average duration, column 2 the average size of change and column 3 the number of observations of such changes.

As can be seen from Table 8.4, only in the case of the UK is there any strong indication of an asymmetry, with interest rate increases tending to be larger, and occurring less frequently (i.e. after a longer duration), than decreases. In the case of the German discount rate, there was a marginal

Table 8.4

		++			--			+-			-+		
		1	2	3	1	2	3	1	2	3	1	2	3
Australia	MM	1.12 ^a	0.74	25	1.13 ^a	0.54	49	1.0 ^a	0.54	18	1.18 ^a	0.54	17
	Discount	22.68	0.49	85	28.6	0.51	86	42.93	0.61	15	47.21	0.54	14
Germany	MM (1980-95)	20.64	0.22	15	22.7	0.25	34	25.83	0.2	24	18.86	0.11	23
	MM (1988-95)	1.24 ^a	0.16	37	1.3 ^a	0.2	22	1.0 ^a	0.09	5	1.25 ^a	0.13	4
Japan	Discount	136	0.7	13	125.9	0.53	24	315	0.5	3	274.3	0.67	3
	MM	1.03 ^a	0.37	113	1.1 ^a	0.42	113	1.13 ^a	0.55	61	1.32 ^a	0.29	60
UK	Discount	7.42 ^a	0.74	15	7.35 ^a	0.55	36	10.67 ^a	0.46	9	13.13 ^a	0.56	8
	MM	42.61	0.81	11	23.55	0.3	84	40.94	0.52	16	37.88	0.56	17
USA	Discount	48.87	1.05	29	38.68	0.61	71	60.4	0.67	16	71.44	1.02	16
	MM	16.88	0.17	88	17.38	0.24	88	20.64	0.18	11	31	0.13	11
	Discount	106.92	0.56	30	94.62	0.51	33	218.22	0.5	9	186.75	0.5	8

^a Only monthly averages available. Durations in months.

indication that ++ were larger than --, but in all other cases there seemed no significant difference between the characteristics of continued increases (++) and decreases (--).⁷ Similarly, though the sample is much smaller, there is no strong evidence (apart again from UK base rates) of any significant differences between the characteristics of +- and -+.

Apart from the UK, continuations of interest rate changes have the same general characteristics, and the same is true for reversals. But there is in most cases a significant difference between continuations and reversals. The duration between continuations is typically lower than between reversals, i.e. after a change of a given sign more working days occur on average before a change in the reverse direction than before a further change of the same sign. This tendency, moreover, is understated by the use of the average duration as the statistical metric. These duration data are seriously skewed, censored on one side where rate changes are measured from the end of one day to the next,⁸ with a very long tail on the other side. We can offset the effects of such a skew (on the average) by taking the median as our metric. If we do so, the data are as shown in Table 8.5.

This reveals clearly enough that a reversal of direction for interest rates generally only occurs after a quite lengthy pause for reflection, and to assimilate new data; whereas continuations tend to follow each other much more quickly. This is so in all cases with respect to discount rates,

Table 8.5: Median duration

	Median duration in declines	Median duration in rises	Median duration between +/— changes	Median duration between —/+ changes
Australian rediscount rate (1974–93)	11	11	21	18.5
Australian call money rate (1984–94) ^a	1	1	1	1
German discount rate (1974–95)	53.5	89	296	336
German OMT rates (1980–8)	13.5	8	6.5	17
Japanese discount rate (1955–95) ^a	5	4	11	10.5
Japanese overnight call rate (1960–95) ^a	1	1	1	1
UK base rate (1974–95)	23	36	45	70
UK band 2 bank bill rates (1981–95)	6	25.5	15.5	15
US FF target rate (1974–9, 1984–92)	9	10	19	30
US discount rate (1962–92)	33	54	163	182.5

^a Monthly average, so duration measured in months; otherwise duration is measured in working days.

Table 8.6 Duration

		> 20 days		> 30 days		> 40 days	
		C	R	C	R	C	R
Australia	MM			monthly data only			
	Discount	47	15	34	11	25	11
Germany	MM	20	19	9	8	7	6
	Discount	37	7	33	7	29	7
Japan	MM			monthly data only			
	Discount			monthly data only			
UK	MM	28	16	24	12	19	9
	Discount	48	26	32	24	25	20
USA	MM	45	16	27	8	20	4
	Discount	48	18	35	18	24	18

but it appears only to be so in Australia and the USA for money market rates. If discount rates are to be altered in a brief space of time, it will be in the same direction. Indeed, the longer the authorities leave interest rates unchanged the less easy it becomes to predict the next direction of change (from the prior history of such changes). Rudebusch (1995, p. 263) demonstrates that once about five weeks (25 working days) have occurred without a change in the Fed funds rate, the probability of a continuation and a reversal become essentially the same.

We examine this same issue by noting all interest rate changes which have occurred more than x days after the prior change, and sorting these into continuations and reversals. This is shown in Table 8.6; this shows clearly that the longer the elapse of time since the last discount rate change, the greater the equality of probability between continuations and reversals. But this syndrome does not appear so marked in the three cases for which we have continuous money market data.

Let me now sum up what has been established in this sub-section. First, the typical pattern of interest rate adjustments by the authorities involves a sequence of relatively small rate changes of the same sign, occurring with relatively small durations between such changes. Only in the UK, where interest rate increases are typically larger and less frequent, is there any real sign of asymmetry between increases and decreases. Third, the longer the delay, after a previous discount rate change, the more likely is the next change to be of equally probable sign (+ or -).

8.2.2 Interest rates and inflation

We now move on to confront the empirical relationship between interest rate changes and inflation in each country. Such a study is made more difficult by several factors. First, how should one measure inflation for this purpose? The inflation rate in any one month is subject to seasonal affects and noise, e.g. owing to indirect tax changes; policy is rarely changed on the basis of one month's data.

Table 8.7

	Series	Source	Period
Australia	CPI (quarterly averages)	IMF IFS 08/95	Q1 1960 to Q2 1995
Germany	CPI (monthly averages)	IMF IFS 08/95	01/1972 to 07/1995
Japan	CPI (monthly averages)	IMF IFS 08/95	01/1960 to 08/1995
UK	RPI (monthly averages)	IMF IFS 08/95	01/1960 to 09/1995
USA	CPI (monthly averages)	IMF IFS 08/95	01/1972 to 09/1995

One method of dealing with both seasonals and noise is to use the year-on-year inflation rate in any one month, measuring the percentage change from the same month in the previous year. But this is, in a sense, a lagging indicator of current inflation, since it gives, in effect, equal weight to inflation in each of the intervening months, and so might be said to give the best estimate of the trend about six months previously. Nevertheless, it is one of the methods that we use. The alternative that we employ is to take the centred average of inflation in month n , from $n - 5$ to $n + 5$ (or in the case of quarters from $n - 2$ to $n + 2$). But this not only fails to remove all seasonals, but also implies the capacity for perfect forecasting over the lead months (quarters).

A technical problem⁹ that arises when we try to compare the timing of changes in inflation and in interest rates is that changes in interest rates generally impinge directly on the RPI (CPI), e.g. via changes in housing costs. Thus an increase in interest rates will have a quasi-automatic effect in raising the RPI (CPI) in the same period. Measures of inflation calculated to remove such interest rate effects were not generally available for our sample countries over our data period; the exercise of trying to recalculate the inflation data ourselves, to exclude interest rate effects from the CPI (RPI), was beyond our capacity. So in interpreting both the charts and the regression exercises, readers should beware that there is a positive bias to the correlation between changes in interest rates and changes in inflation in the same month. The actual series of inflation that we use in each case¹⁰ is shown in Table 8.7.

A third problem that we face is one of simultaneous causation. Insofar as the authorities can observe, or predict, the policy-unchanged deviation of inflation, π , from its desired level, π^* , at $t = 0, 1, 2, \dots, n$, and can vary i now in such a way as to reduce that deviation, that deviation ($\pi - \pi^*$) would be reduced. Consequently, the correlation between i_t and $(\pi - \pi^*)$, $i = 0, 1, 2, \dots, n$ would be changed, generally reduced, and at the limit, with perfect stabilization policy, brought to zero. While this latter would not be a good description of events in any country (though on this criterion Japan has recently done best), there is a need to be aware that the estimated relationships may be biased, probably downwards.

A fourth problem that we face is how we should estimate π^* , the desired level of inflation. Going back over the historical record, and trying to extract contemporary policy statements would have been too time consuming for the purposes of this exercise. So, I took a simple, but

Table 8.8: Sum of deviations from desired inflation

	1973-95	1973-84	1985-95
Australia	2142	1537	605
Germany	478	336	142
Japan	933	766	167
UK	1787	1461	326
USA	940	780	160

arbitrary, measure; I took π^* , the desired level of inflation, to be a *constant* (at 2.5 per cent over the whole period).¹¹

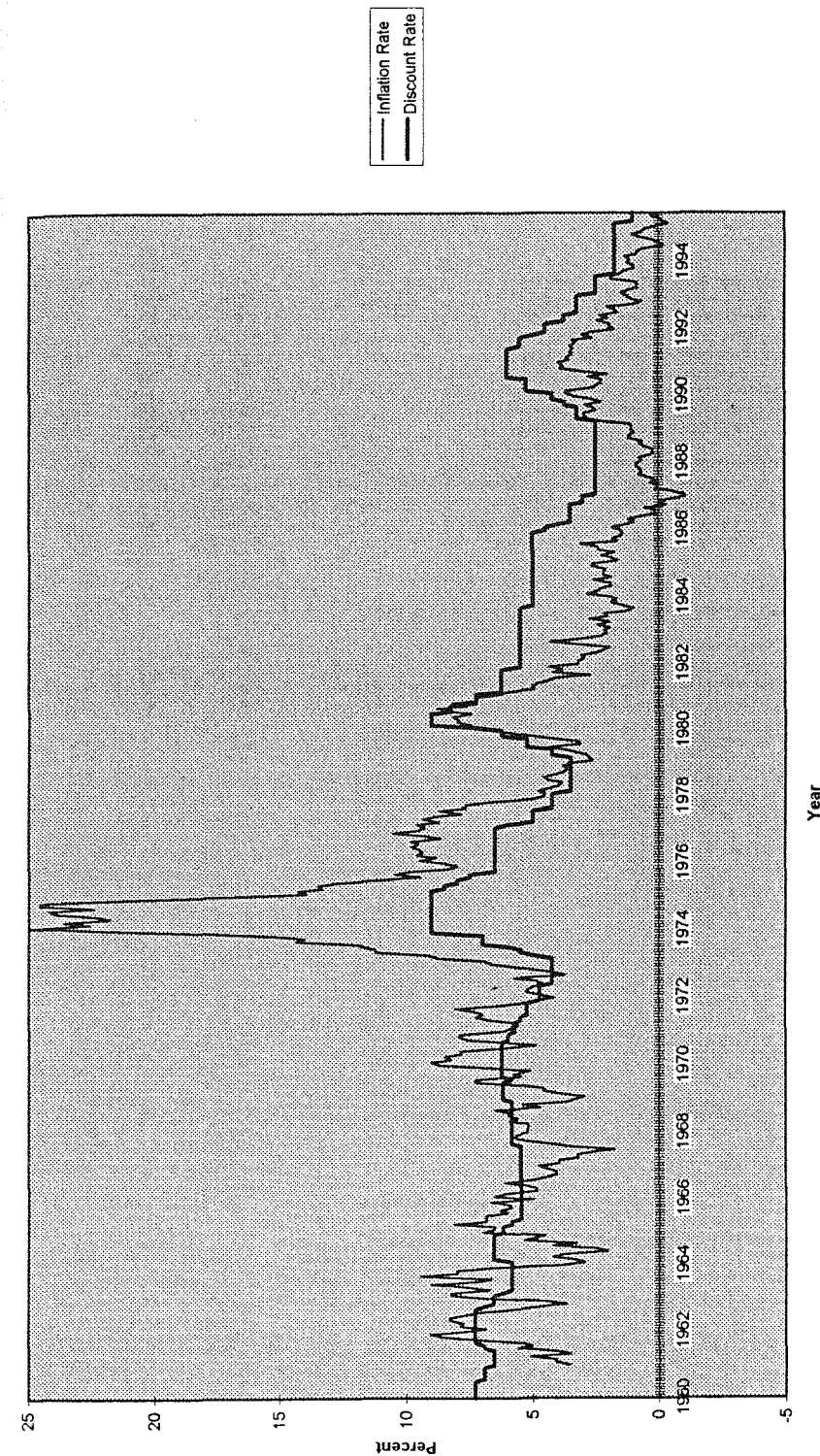
We can now calculate some of the characteristics of such undesired inflation/deflation, concentrating on the case where π^* is held fixed at 2.5 per cent throughout. Diagrams of these are shown in Figures 8.12 to 8.16. In Table 8.8, we first estimate the total sum of deviations, without regard to sign, as a measure of overall success, over the period 1973-95, and the two half-periods 1973-84 and 1984-95.

The clear 'winner' is Germany, but the scale of its comparative success was much greater in the first period, when it coped much better than any other country with the inflationary pressures of the oil shocks and the other disturbances in the 1970s. In the second half-period, Japan and the USA do just about as well.

But visual inspection of the graphs suggests that deviations from the (arbitrarily chosen) desired level were smaller in Germany, but just about equally, or more, persistent than in other countries (though such visual comparison is complicated by the differing scales). We tested for this by running the following equation for each country:

$$\Delta\pi_t = \alpha + \beta_1\pi_{t-1} + \beta_2\Delta\pi_{t-1} + \beta_3\Delta\pi_{t-2} + \beta_4\Delta\pi_{t-3} + u_t$$

This is, of course, an augmented Dickey-Fuller test for stationarity. This equation may help to give us three bits of evidence: first, the strength of the tendency to revert back to the desired level (β_1 , which should be negative); second, the strength of continuing autocorrelation, i.e. the values of $\beta_2, \dots, \beta_4$, and finally the relative size of shocks, the size of the u residuals. In Table 8.9 we use the centred data for inflation, but the year-on-year data results (available on request) are broadly similar. The results are shown for the two half-periods, 1973 M5 to 1982 M12 (116 observations), and 1983 M1 to 1995 M2 (146 observations) separately; the Australian data are, however, quarterly (36 and 48 observations, respectively).

**Figure 8.12** Japanese monthly inflation and discount rates, 1960-1995

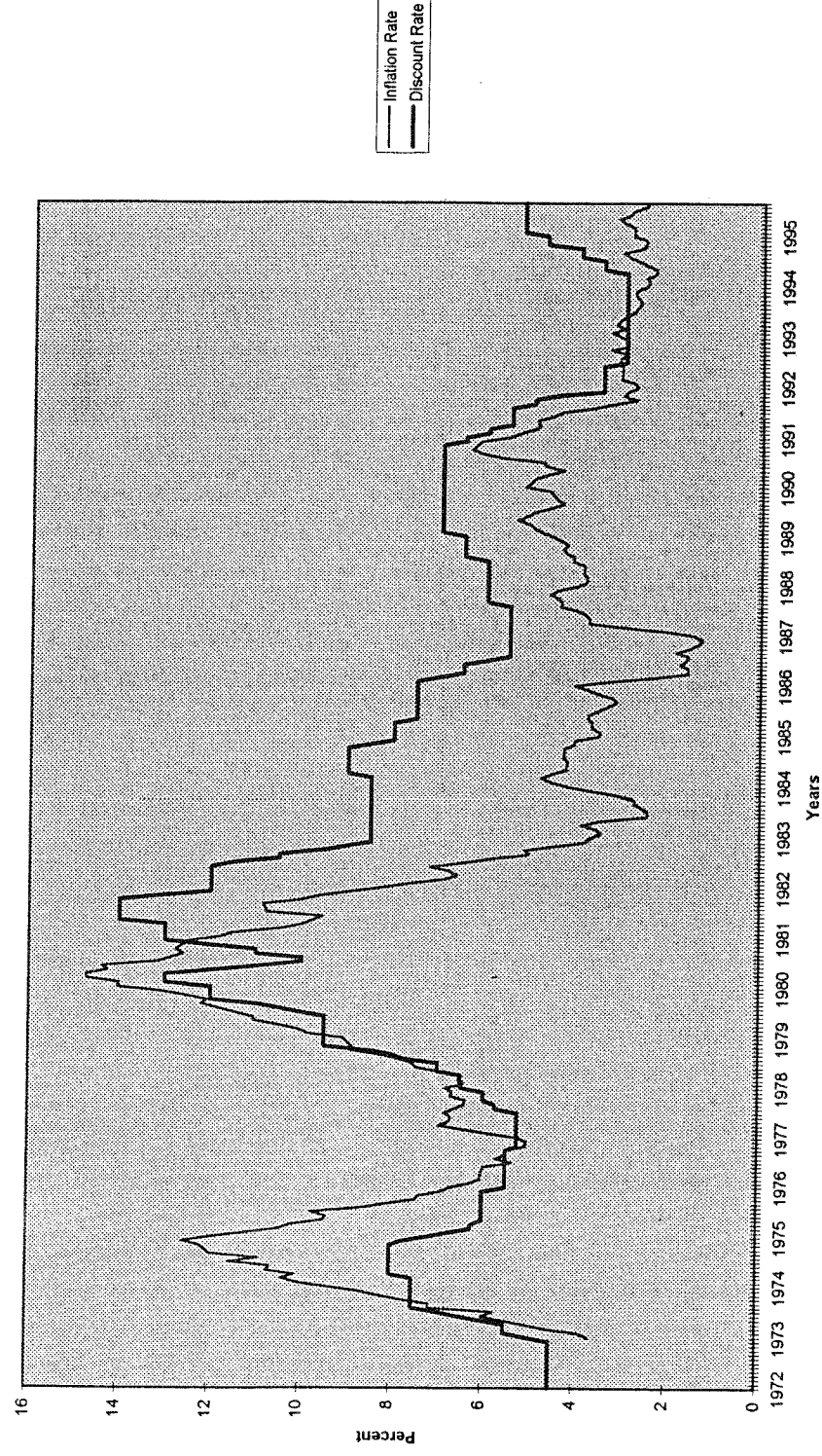


Figure 8.13 US monthly inflation and discount rates, 1972–1995

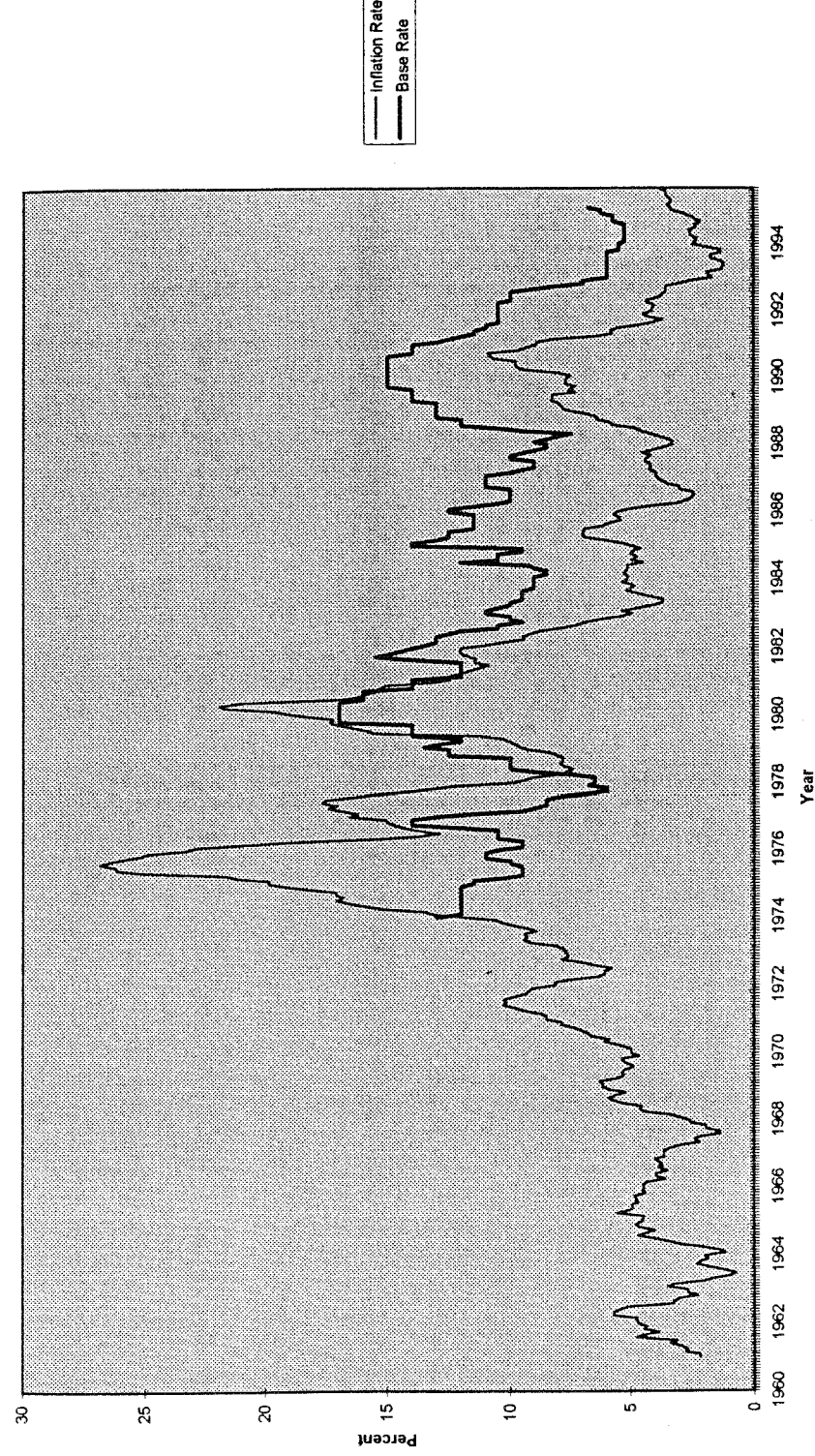


Figure 8.14 UK monthly inflation and base rates, 1960–1995

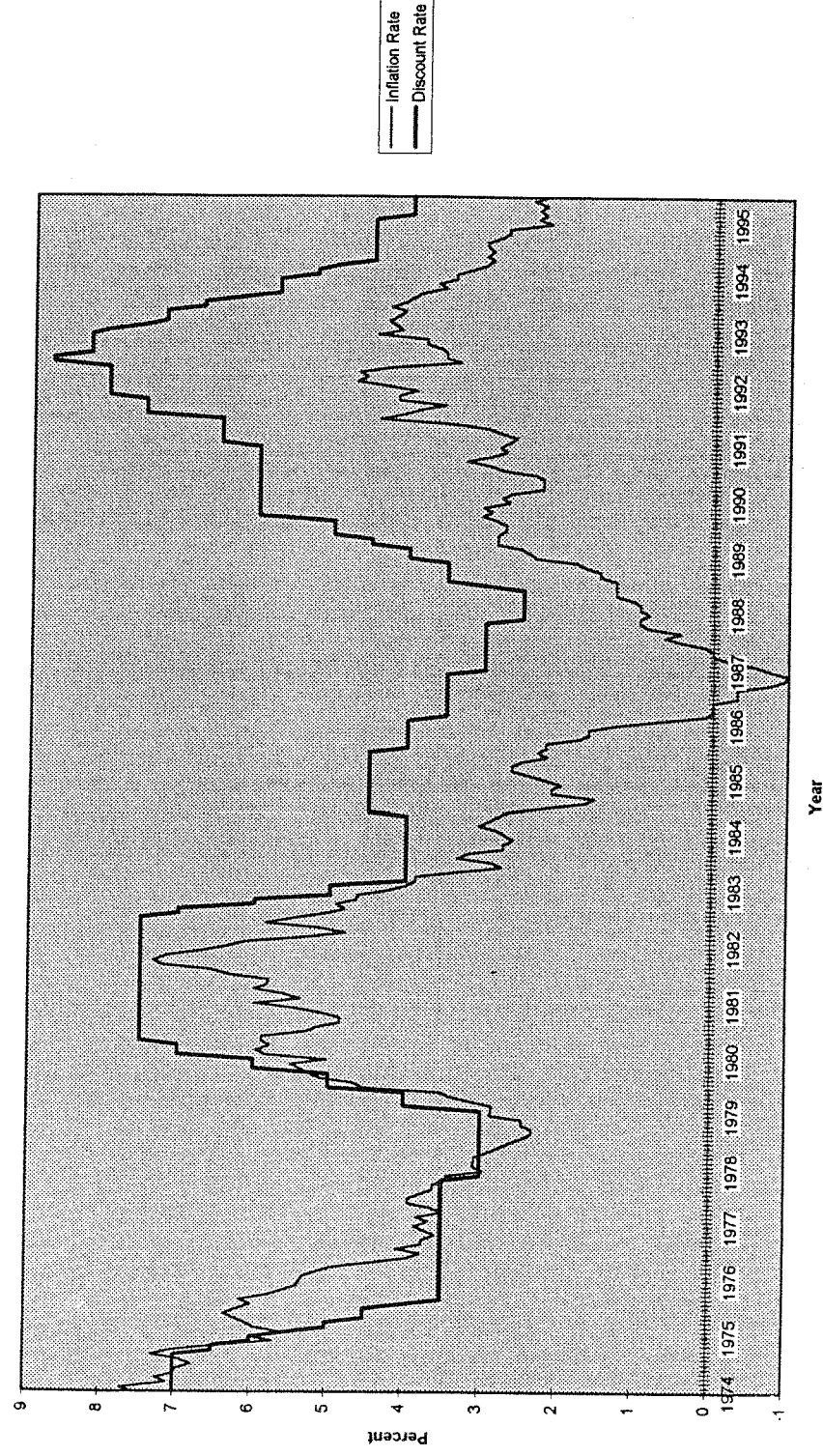


Figure 8.15 German inflation and discount rates, 1974–1996

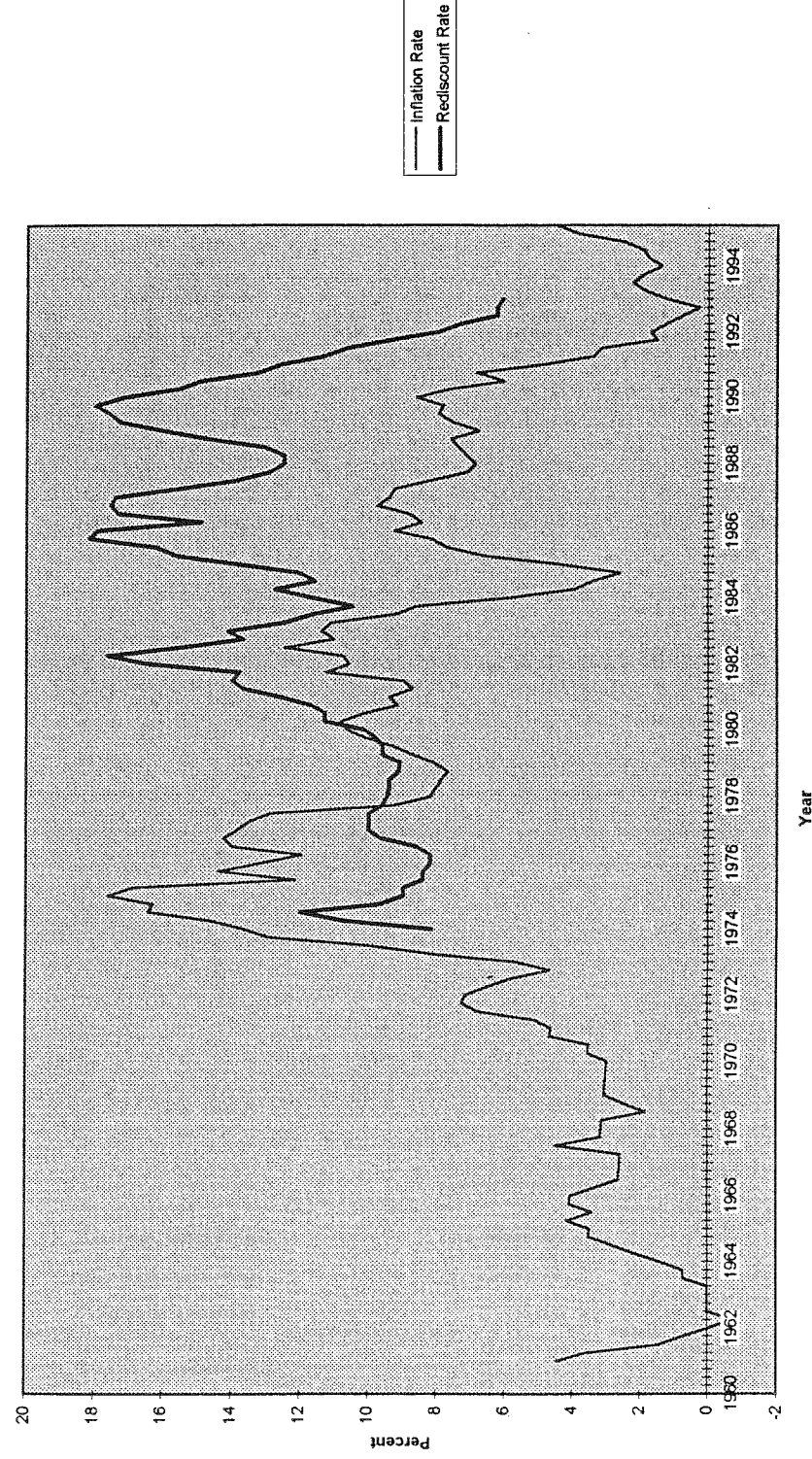


Figure 8.16 Australian quarterly inflation and rediscount rates, 1960–1995

Table 8.9 ADF equations

	β_1	$\Sigma\beta_2 + \beta_3 + \beta_4$	Σu^2
<i>1973 M5 to 1982 M12</i>			
Australia	-0.22	0.545	48.8
(qtrly data)	(-2.57)		
Germany	-0.05	0.236	16.0
	(-1.81)		
Japan	-0.04	-0.001	192.2
	(-1.54)		
UK	-0.06	0.497	174.7
	(-2.23)		
USA	-0.02	0.586	18.7
	(-1.18)		
<i>1983 M1 to 1995 M2</i>			
Australia	-0.13	0.697	34.8
(Qtrly data)	(-2.97)		
Germany	-0.04	0.050	14.1
	(-1.67)		
Japan	-0.14	-0.468	61.4
	(-2.46)		
UK	-0.05	0.008	55.6
	(-1.85)		
USA	-0.07	0.451	9.1
	(-2.84)		

The coefficient β_1 indicates the extent to which pressure has been placed on the series in question to revert to a mean level. The critical level at which we can reject the unit root hypothesis is -2.75 . If we hypothesize that such pressure for mean reversion comes from the authorities, then this table provides an entirely different perspective. The countries most forcefully counteracting inflationary shocks are Australia and the UK in the first period, and Australia, Japan and the USA in the second. Germany made the least effort in the second period, and together with the USA and Japan made relatively little effort in the first period. Indeed, over these periods you could not reject the hypothesis that inflation in Germany was unit root, i.e. there was no tendency towards mean reversion.

How, then, did Germany perform so well overall? By this measure, it faced smaller shocks, (lowest Σu^2), as did the USA, and when a shock came, there was only weak persistence. By contrast, both Australia and the UK suffered from both large shocks and strong persistence. On the overall measure in Table 8.8, Australia did worst, but by this measure (Table 8.9), it struggled the hardest of all to respond to extremely adverse conditions. Japan also had very large inflationary shocks, but no persistence. The USA faced relatively small shocks, and moderate persistence.

No doubt the data in Table 8.9 can be interpreted in different ways, and differing data series and periods would give different answers. But the perspective on inflation control presented here is markedly different from

that which comes from the standard approach, represented by Table 8.8. Is it a valid alternative viewpoint? Does it help to explain, or is it simply another way of expressing, the otherwise paradoxical finding that Germany has had the best overall inflationary record, yet appears at the same time to be the chief exponent of 'too little, too late', and interest rate smoothing? We shall revert to this question shortly.

Next we shall turn from consideration of the characteristics of inflation to an examination of the relationship between inflation and interest rates. These are first shown graphically, for the relationship between discount rates and inflation¹² (year-on-year), 1974-94, in Figures 8.17 to 8.21 for our five countries. The first point to note is that there is no sign of any sizable lead of interest rates over inflation in any of these countries, despite the fact, discussed earlier, that the year-on-year measure of inflation is a (six-month) lagging indicator. We formalized this by examining at what lead/lag between the two series the simple correlation was maximized. The results are shown in Table 8.10.

Table 8.10

Rate lags					Rate leads			
-4	-3	-2	-1	0	+1	+2	+3	+4
	JM		JD	GM	AM		UKD	
	USD			GD	AD			
					UKM			
					USM			

The Australian data are quarterly; monthly for the other countries.
M is money market rate; D is discount rate.

The relatively tardy adjustment of the US discount rate is consistent with accounts of the use of that instrument in policy operations there. There is some evidence here that the UK and Australia are prepared to take action rather more quickly in response to inflationary threats than the other three countries. But the speed of response remains pitifully slow. If we replace the year-on-year rate of inflation with our measure of the current centred rate of inflation, then all the relationships become lags, subject to a doubt about Germany, whereby the discount rate and money market levels lag the inflation rate. On this basis, Australia, the UK and Germany become the relatively speedier respondents to current inflationary pressures, whereas the USA and Japan have taken longer to react (see Table 8.11).

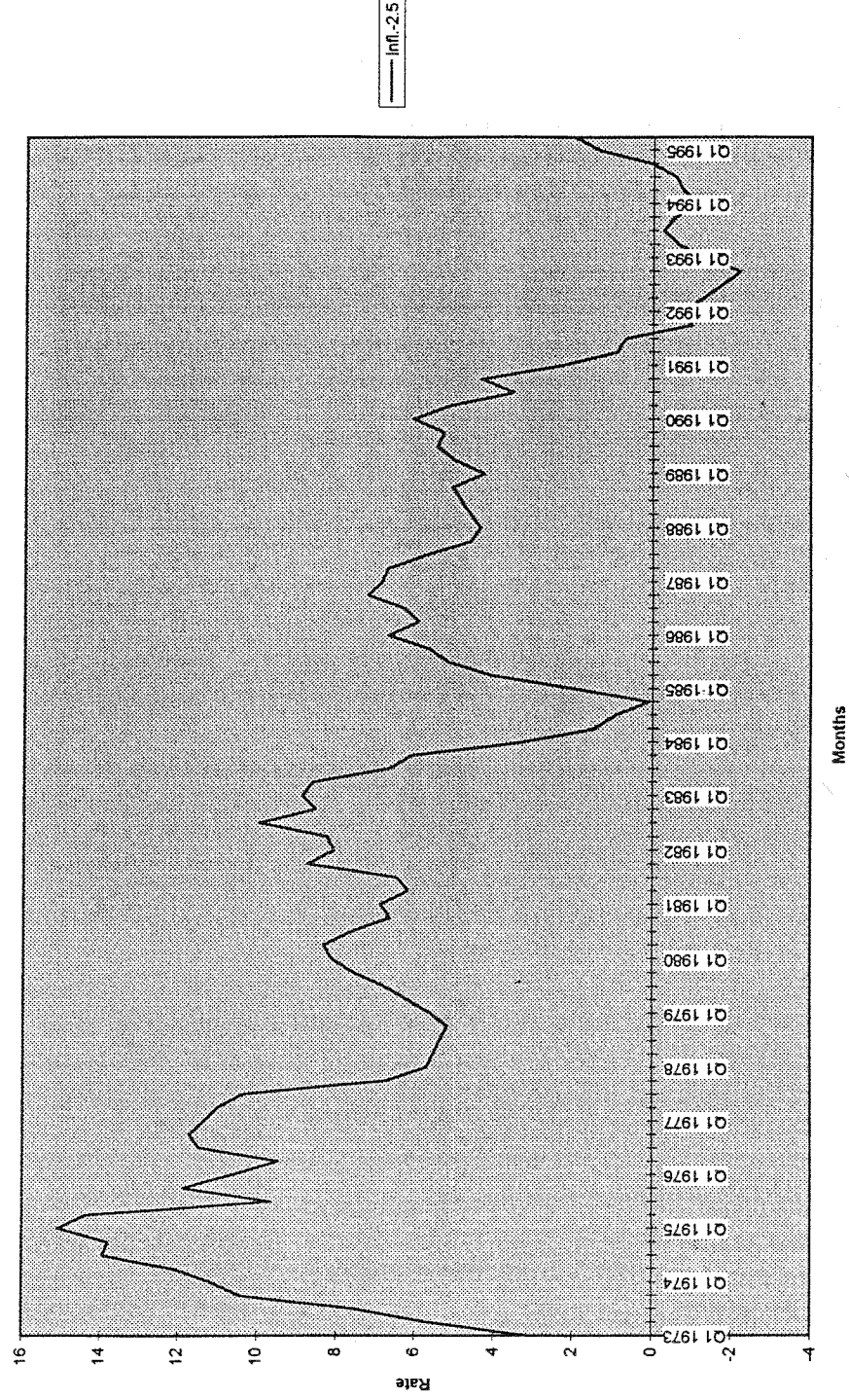


Figure 8.17 Australia: inflation rate — 2.5 per cent

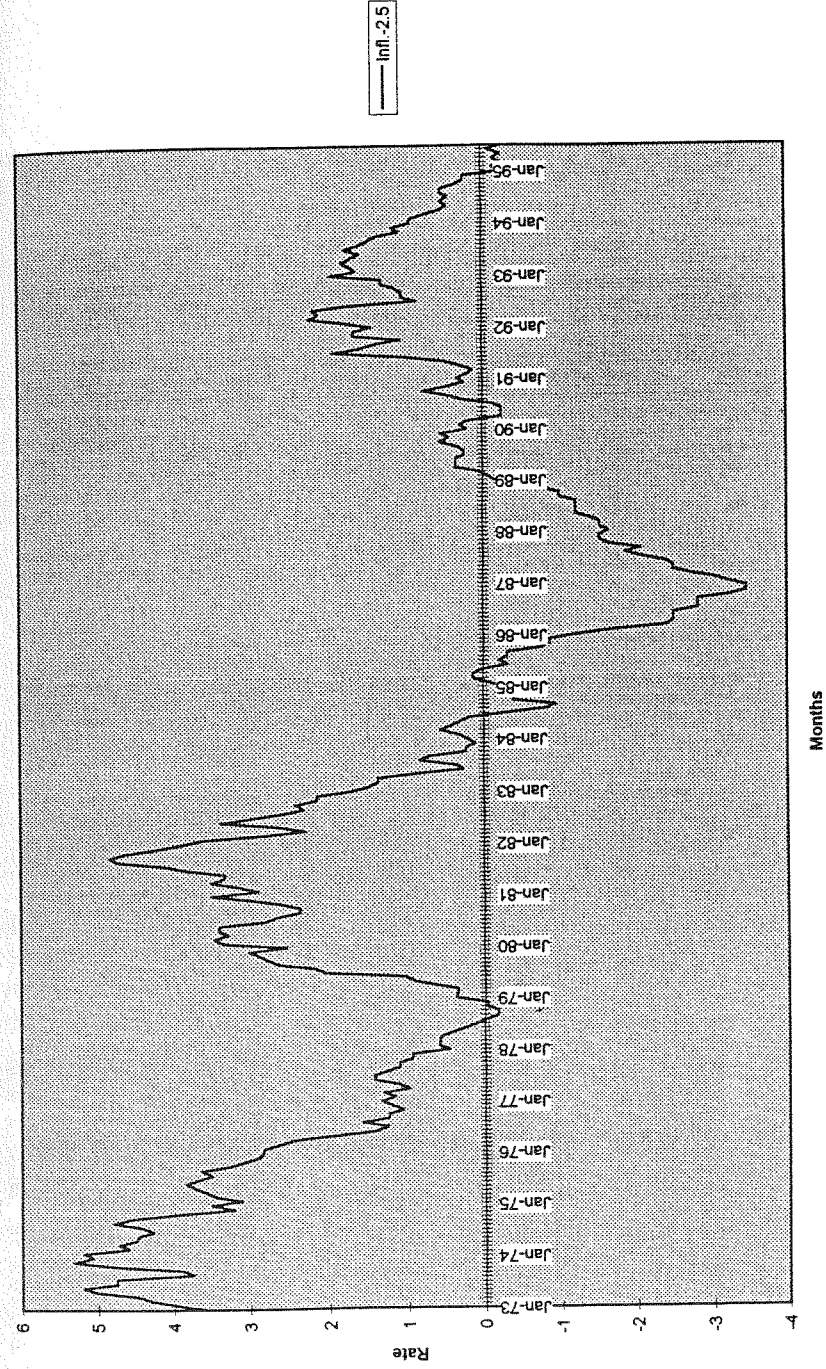


Figure 8.18 Germany: inflation rate — 2.5 per cent

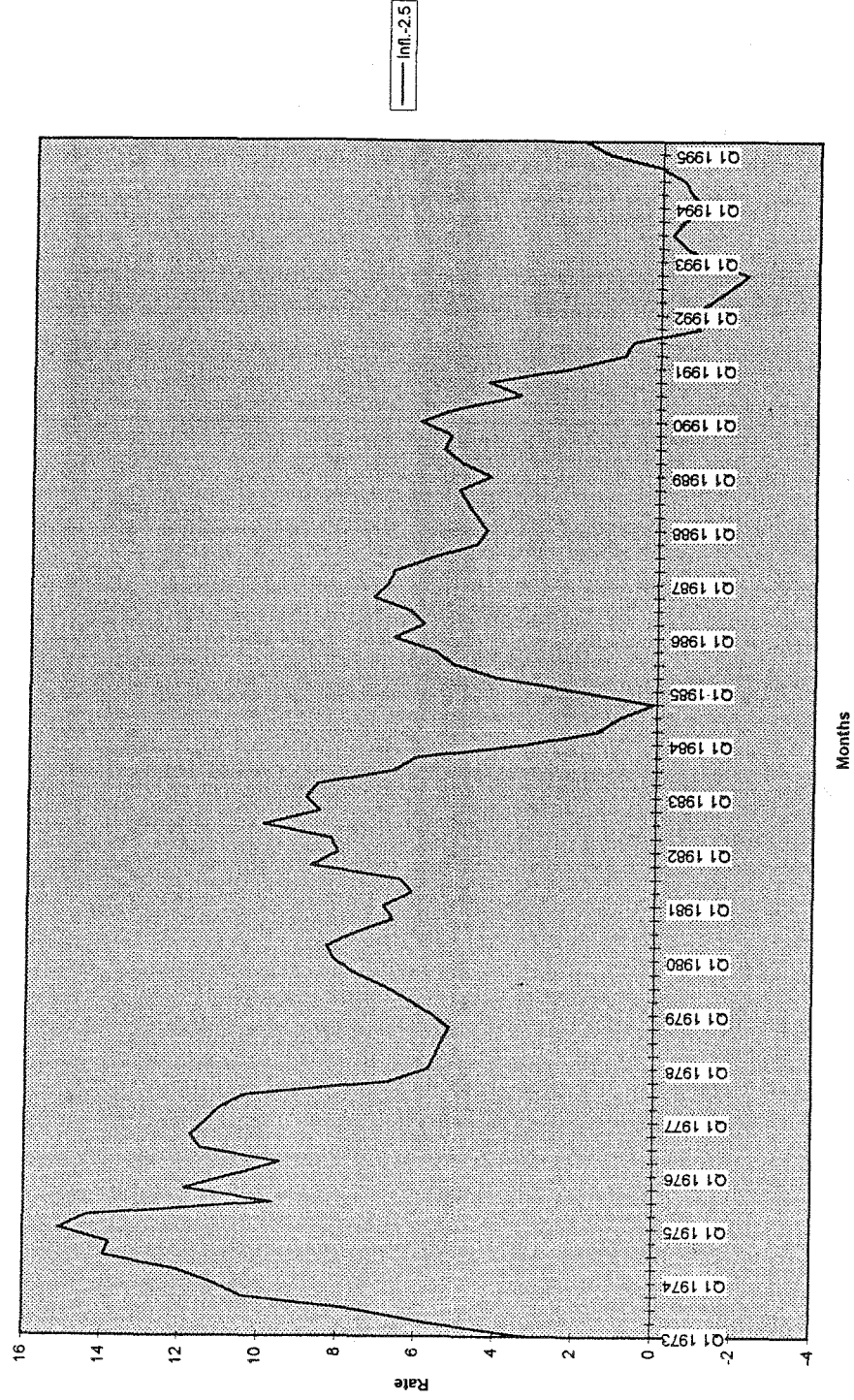


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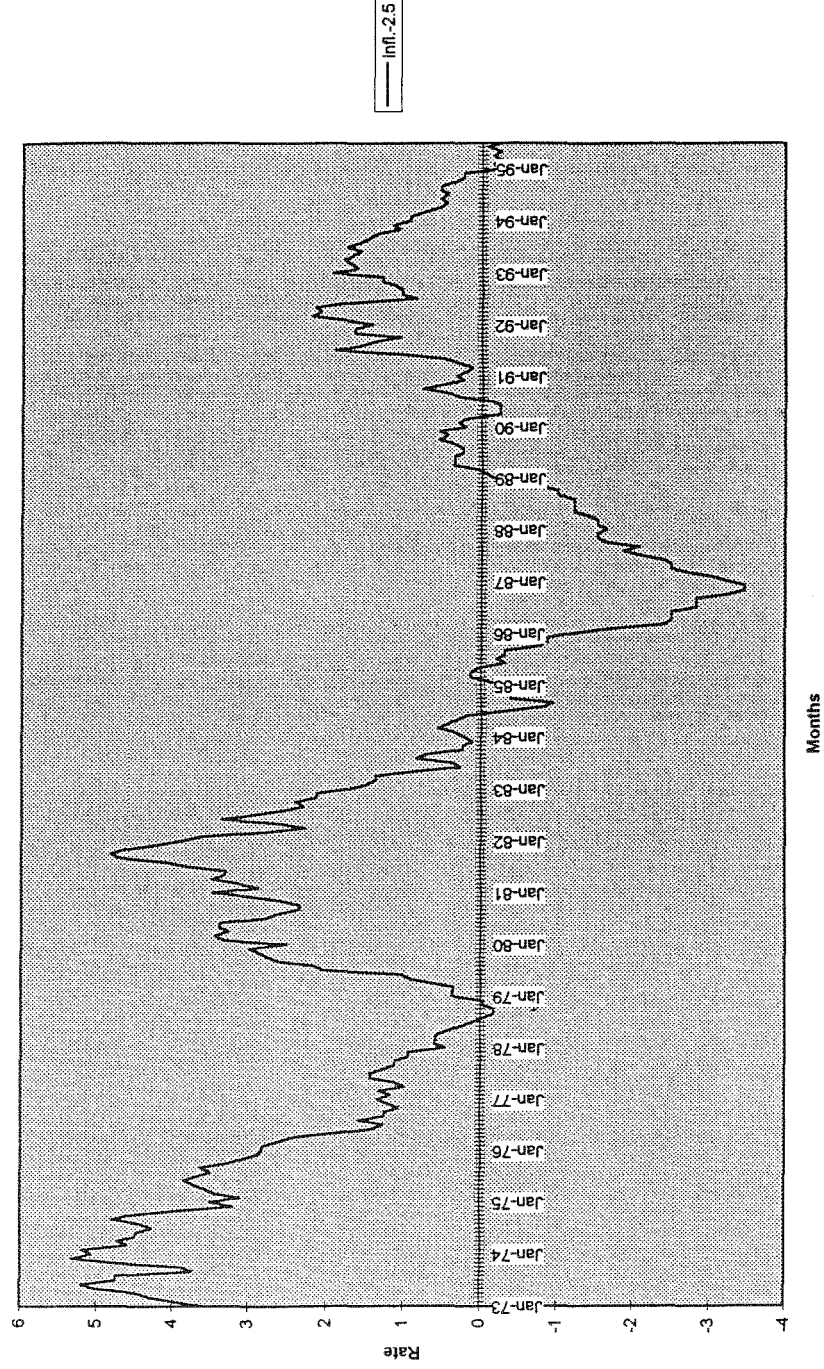


Figure 8.18 Germany: inflation rate — 2.5 per cent

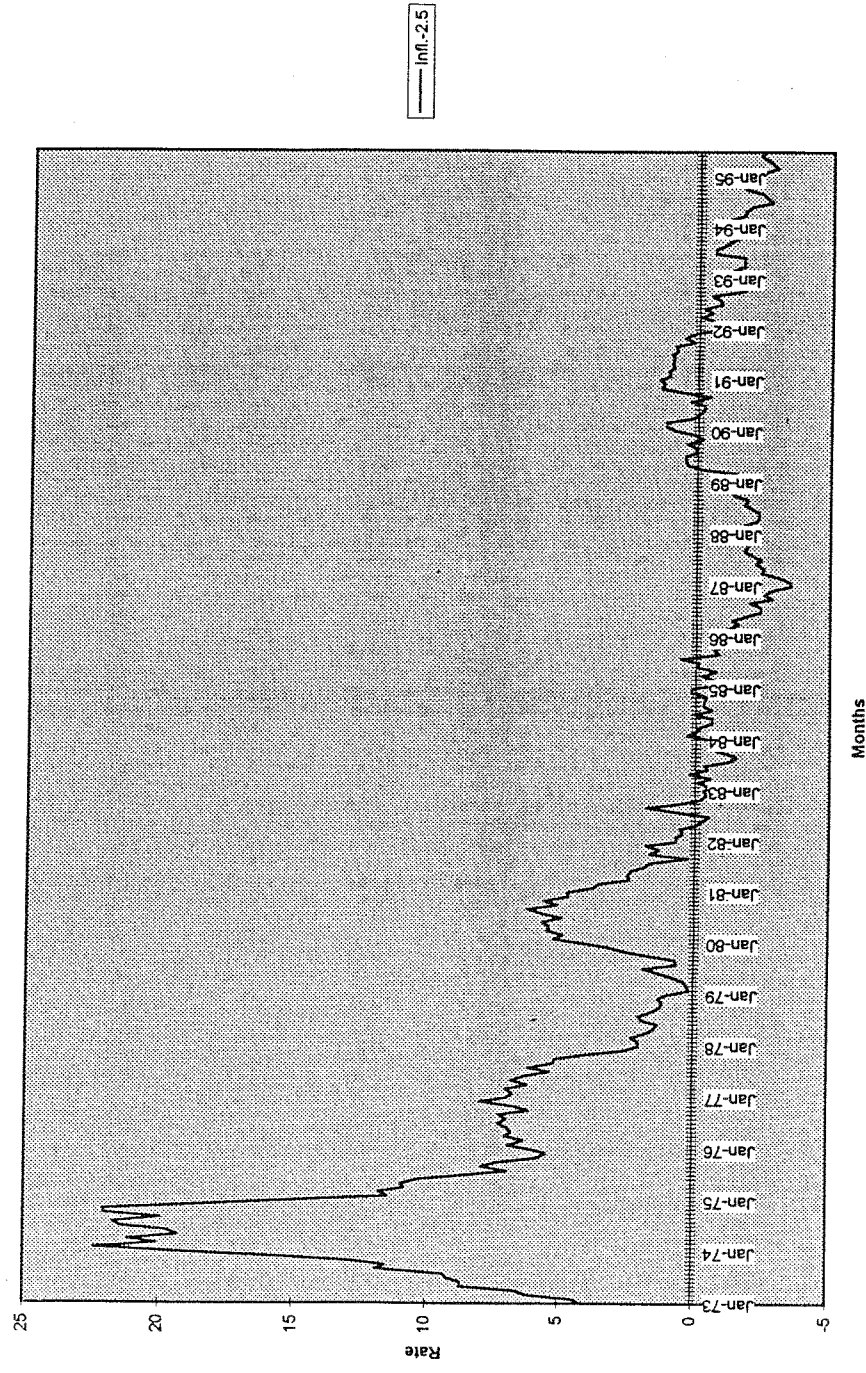


Figure 8.19 Japan: inflation rate - 2.5 per cent

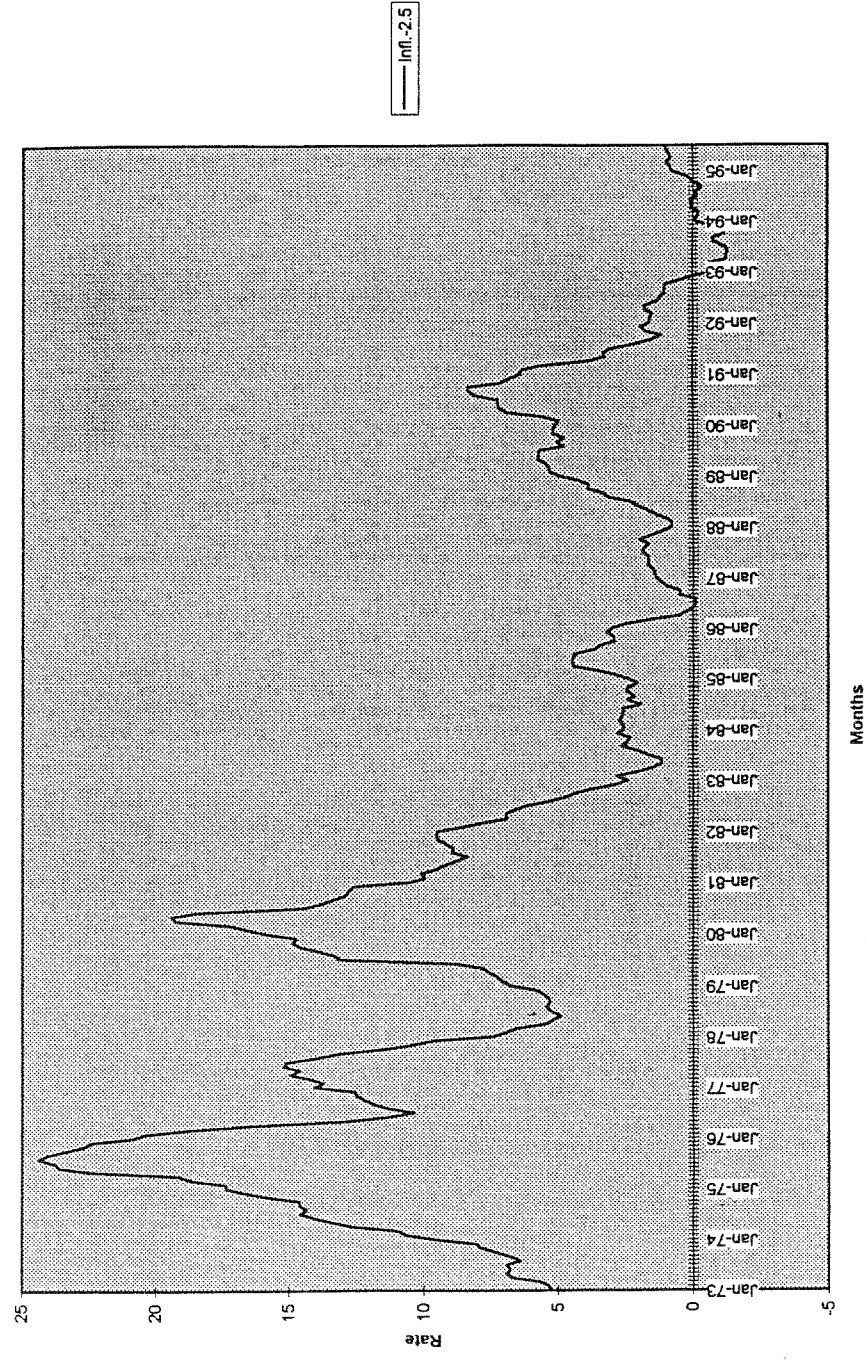


Figure 8.20 UK: inflation rate - 2.5 per cent

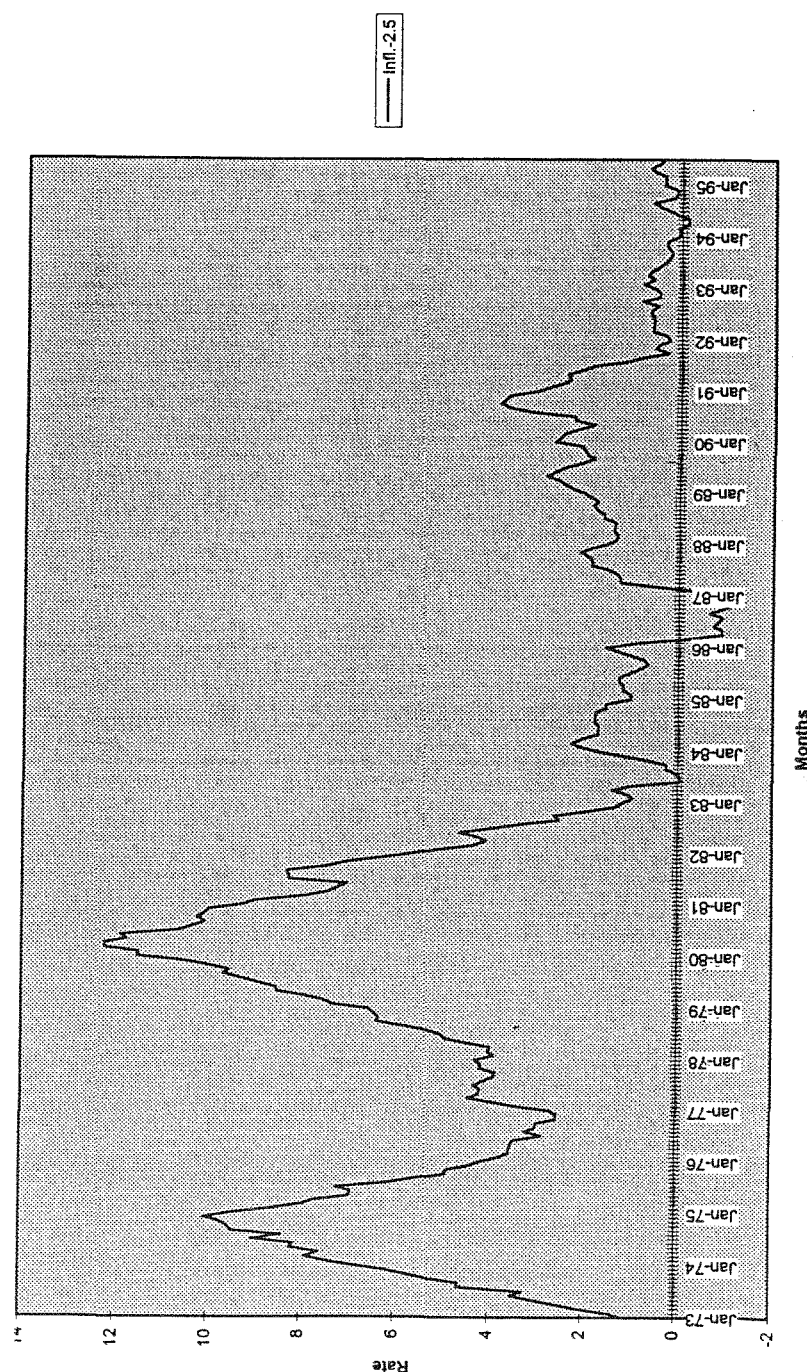


Figure 8.21 US: inflation rate - 2.5 per cent

Table 8.11 Length of lag of rate change behind inflation

	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0
USD													
USM													
JM													
JD													
UKM													
UKD													
AM													
GD													
GM													
AD													

Australian data are quarterly; monthly in other countries:

German data are not monotonic; there is a slightly higher peak correlation with German interest rates leading inflation by 2 (M) and 5 (D) months respectively. Our inclination is to dismiss this as spurious given the weight of other evidence.

M is money market rate; D is discount rate.

Again we can examine the relationship between changes in inflation (both lead and lag), together with the contemporaneous deviation between inflation and its desired level (δ), and changes in money market interest rates. If the authorities could both predict and respond to their forecast of inflation, then, given the lags in the operation of monetary policy, one would hope that changes in interest rates at time t would react to *future* changes in inflation (subject to the qualification about simultaneous bias already mentioned). For this exercise we use quarterly data. We regressed the change in discount rates in the current quarter against the change in inflation in the subsequent four quarters, ($t + 1$ to $t + 4$), in the current quarter, and in the prior four quarters, ($t - 1$ to $t - 4$), and also against the current level of inflation. We ran the exercise using both measures of inflation, current centred and also year on year. The results are shown in Table 8.12.

What this suggests is that the authorities respond positively, but quite weakly, to deviations in the *level* of inflation (above its desired level). Taking the average response of the five countries, at 0.07 (0.05 year-on-year inflation), should the level of inflation be 1 per cent higher than desired, it would take the authorities about 14 (20) quarters, 3.5 (5) years, to raise interest rates by 1 per cent to compensate. Note that again the German interest rate response to the level of inflation is the lowest.

The authorities appear to react much more firmly to changes in interest rates. In three cases (USA, UK and Australia) in Table 8.12, with the longest lead ($t + 4$ quarters), with centred inflation data, there was some sign of a significant positive relationship. We tested whether this could have been owing to truncation of the number of leading quarters, but when we added two further lead quarters, that positive relationship even strengthened marginally in each case (the coefficients on the leads at $t + 5$ and $t + 6$ were negative). At present we have no good explanation for this. For the rest, using the centred inflation data, there was, in three cases (Japan, the UK and the USA), some evidence of a move in interest rates when the inflation rate led by one quarter. This lead was, not surprisingly, even stronger when we used year-on-year data (recall again that this measure of inflation incorporates a built-in lag of about two quarters); in this latter case, there were fairly clear signs of a one quarter lead, of interest rates responding to a (forecast) acceleration in inflation in the subsequent quarter, and perhaps a longer lead in Japan. The overall

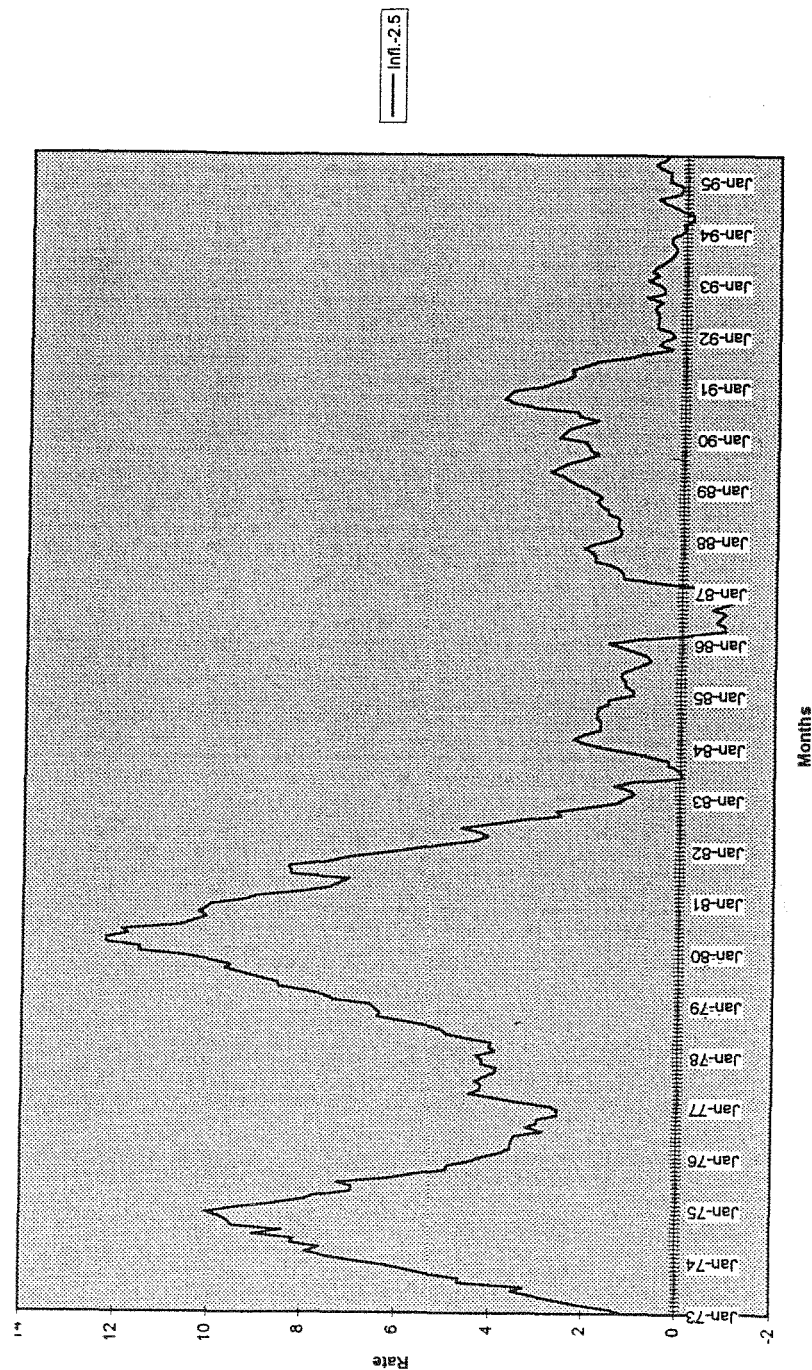


Figure 8.21 US: inflation rate - 2.5 per cent

Table 8.11 Length of lag of rate change behind inflation

	-12	-11	-10	-9	-8	-7	-6	-5	-4	-3	-2	-1	0
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USM													
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JD													
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GD													
GM													
AD													

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Table 8.12 Response of interest rates to inflation

	Changes in Inflation										Level of Inflation
	Leading					Lagging					
	+4	+3	+2	+1	0	-1	-2	-3	-4	0	
<i>With centred inflation</i>											
Australia	0.27 (1.86)	0.14 (0.95)	0.14 (0.98)	0.07 (0.47)	0.06 (0.39)	0.25 (1.96)	0.18 (1.49)	0.12 (0.93)	0.11 (0.92)	0.09 (1.92)	
Germany	0.02 (0.16)	0.04 (0.43)	0.11 (1.08)	0.03 (0.24)	0.18 (1.67)	0.18 (1.79)	0.14 (1.38)	0.17 (1.69)	0.16 (1.51)	0.04 (0.78)	
Japan	0.01 (0.20)	-0.01 (-0.20)	-0.02 (-0.24)	0.17 (2.68)	0.22 (3.78)	0.16 (3.00)	0.24 (4.54)	0.18 (3.63)	0.01 (0.13)	0.04 (1.26)	
UK	0.14 (1.80)	-0.00 (-0.03)	0.03 (0.26)	0.34 (3.04)	0.08 (0.71)	0.08 (0.77)	0.16 (1.92)	0.07 (0.79)	-0.03 (-0.29)	0.08 (1.80)	
USA	0.26 (2.20)	-0.03 (-0.24)	-0.22 (-1.80)	0.18 (1.46)	0.09 (0.79)	0.12 (0.99)	0.03 (0.27)	0.06 (0.50)	0.12 (1.10)	0.10 (3.02)	
<i>With year on year inflation</i>											
Australia	0.21 (1.41)	0.18 (1.19)	0.05 (0.35)	0.15 (1.15)	0.15 (1.01)	0.18 (1.35)	0.10 (0.79)	-0.12 (-0.96)	-0.08 (-0.62)	0.08 (1.62)	
Germany	-0.09 (-0.85)	0.08 (0.73)	0.15 (1.44)	0.22 (2.10)	0.06 (0.49)	0.23 (2.15)	0.12 (1.22)	0.24 (2.53)	0.07 (0.70)	-0.01 (-0.31)	
Japan	0.11 (1.54)	0.08 (1.10)	0.16 (2.19)	0.28 (3.85)	0.21 (2.58)	0.10 (1.65)	-0.12 (-0.22)	0.10 (1.79)	0.02 (0.39)	0.05 (2.25)	
UK	0.15 (1.20)	0.19 (1.58)	-0.08 (-0.65)	0.31 (2.62)	0.17 (1.28)	0.12 (1.10)	-0.31 (-2.64)	0.18 (1.63)	0.08 (0.72)	0.05 (1.45)	
USA	-0.17 (-1.58)	0.24 (2.34)	-0.07 (-0.66)	0.38 (3.52)	0.07 (0.58)	-0.03 (-0.33)	-0.03 (-0.32)	0.14 (1.39)	0.01 (0.09)	0.07 (2.57)	

Table 8.13

	Centred data		Year on year data	
	Sum of coefficients 5 quarters on changes	Coefficient on level	Sum of coefficients 5 quarters on changes	Coefficient on level
Australia	0.72	0.09	0.64	0.08
Germany	0.83	0.04	0.87	-0.01
Japan	0.97	0.04	0.83	0.05
UK	0.73	0.08	0.71	0.05
USA	0.48	0.10	0.59	0.08

impression is that a 1 per cent change in inflation will provoke a responsive increase in nominal interest for about four or five consecutive quarters (from $t + 1$ to $t - 3$ or t to $t - 4$) of about 0.15 per quarter and that this reaction is broadly similar in all countries. There are some slight indications that in countries where the response to inflation levels is highest, the reaction to changes is lowest. Thus, we have Table 8.13.

Given that monetary policy affects inflation only with a considerable lag, as many other studies have attested, the fact that interest rates appear to respond primarily to current and prior, and only slightly to short future, changes in inflation gives substance to the claim that they are varied 'too late'. Our earlier finding that, once interest rates do change direction, this is usually continued in a sequence of a number of relatively small step changes, gives substance to the claim that they are varied 'too little'.

Central banks should plead guilty to the charge of 'too little, too late'. Does this occur because their forecasting ability is no good; because they are actually trying to control some other variables besides, or instead of, giving inflation primacy; or because they are (politically) constrained from responding to forecasts and are forced to react to actual, historical data?

8.2.3 Why was Germany so successful?

One of the more interesting findings of this empirical section is that tendencies towards 'too little, too late' do not seem to be closely related to overall counter-inflationary success. As calibrated by our various metrics, Germany and the USA have been most prone to this syndrome (slow and steady as she goes), while Australia and the UK have been the most aggressive in response to inflationary threats.

If this empirical finding should be accepted, there are a variety of possible explanations. The first hypothesis is that inflationary shocks have been much more extreme in the UK (and in Australia) than in Germany and the USA. The central banks in the UK and Australia have, indeed, been more determined/aggressive in trying to overcome such shocks, but they have simply not been able to cope with a more adverse conjuncture. The US experience seems to fall into two separate states, with much smoothing

and a lack of success until 1979, and more aggression and much more success thereafter. The central bank in Germany may not have done 'better' in any objective sense. It may simply have been dealt a much easier hand to play; the evidence in Table 8.9 is consistent with this hypothesis.

One important comment made at the conference by Professor Alex Cukierman, but generally supported, was that the degree of persistence in the sample countries of inflationary shocks could not be taken as exogenous. It was possible that the apparent low persistence of inflationary shocks in Germany could have been owing to structural factors, among which the role of the authorities could have played an important part. Another comment was that German reunification had been a major shock, so that any suggestion that Germany had suffered significantly smaller inflationary shocks than the other countries in this group needed to be viewed sceptically. While the empirical results in Table 8.9 were interesting, partly because they did provide a different perspective, there was little willingness among those present at the conference to accept at face value the suggestion that the Bundesbank may have done better at inflation control simply because it had faced a much more benign conjuncture.

A second hypothesis is that, at some prior date, the central banks of Germany (and Japan?) built up a reputation for credibly holding future inflation to a low stable level. Hence a given nominal interest rate increase in these countries implies a much larger increase in real interest rates than the equivalent in the UK or Australia. So, even though Germany appears to react less aggressively than the UK in nominal terms, it is actually being more aggressive in real terms.

Given the importance attached to the concept of credibility, it is surprising that so little comparative cross-country work has been done on this. Weber (1991) did an exercise to study the comparative credibility of monetary announcements in EMS countries. But he had no direct evidence, or data series, on expectations, and so had to construct artificial series. As both his discussants, R. Baldwin and M. Obstfeld, pointed out, his results may well have been largely driven by the errors in such construction.

Now that survey data on inflationary expectations are available in more countries and over a longer period, it should be possible to examine how such expectations react to previous events. Thus, one might expect that in countries with more credible central banks an upturn in current, and past, inflation should have less effect on expectations of future inflation. Similarly, an unanticipated rise in interest rates in countries with more credible central banks should reduce expectations of inflation by more (than in less credible countries). A VAR exercise relating interest rates and expected and actual inflation (across countries) might be instructive. Such an exercise was again beyond my capacities here, but I may be able to turn to it in future.

A third hypothesis is that the very predictability of long continued sequences of money market rate changes allows longer rates, e.g. of six months to one year, to adjust much more rapidly, and it is the latter which matter. This is the provisional Goodfriend-Rudebusch hypothesis. Thus

the continuing sequence of small steps is a kind of game played by the authorities and understood by markets. But, if so, what is the point of this game? Why not just raise interest rates on money markets by more at the outset and then leave them constant? Anyway, this hypothesis implies that longer-term, six-month to one-year rates adjust sharply in response to a (prospective) change of direction in money market rates,¹³ and then decreasingly to further steps in the same direction in money market rates. This is very likely so to some extent; papers by Miles and Rudebusch indicate that this could well happen in the UK and the USA. I do not know of any systematic cross-country work on this hypothesis.

On this, Miles (1994) writes:

But for both the UK and US the responsiveness of longer bond yields to changes in short rates is substantially higher at turning points than at other times; when short term rates are increased after a sustained period of easing in monetary policy, or when policy is relaxed after a series of interest rate increases, the impact upon longer term interest rates is much larger than usual. Table [8.14] shows the change in US 10 year bond yields around the last seven turning points in US monetary policy. The table measures the response of the yields on the 10 year benchmark bond over a period from 2 days before the change in the federal funds rate to 2 days after; Cook and Hahn find that expectations and learning effects around the time of changes in monetary policy are significant so we allow a couple of days either side of the actual change in official rates for their full impact to be felt. On average long bond yields in the US respond by around 100% of the change in short rates at turning points. Dale (1993) tested for enhanced effects of monetary policy changes at turning points using a series of turning point dummies in regressions of the change in longer rates on the change in short rates set by the Bank of England. For all maturities those dummies were highly significant; at the 10 year horizon bond yields were estimated to change hardly at all at non turning points but to respond significantly when there was a switch in the direction of interest rate changes.

Because of our interest in the German experience, we attempted to do a similar exercise for Germany. We obtained data from the Bundesbank on German bund yields with a remaining maturity of between eight and fifteen years. These data gave weekly observations from 7 January 1982, when this data series began, until 6 November 1987. Thereafter the data were daily, covering the period until 17 November 1995, when our data series ended. Following the previous example of Cook and Hahn (1989) we calculate the change in bund yields over the week in which the discount rate changed, for the period with weekly observations only, and from two working days prior (i.e. excluding weekends and holidays) to two working days afterwards over the period when we had daily data.

The hypothesis which we were testing was that the change in bund yields would be much larger when accompanying a reversal in direction, since markets would then anticipate a series of subsequent similar small moves in discount rates, than otherwise.¹⁴ The significance and value of the test was, however, much reduced by the fact that there were only five reversals during our data period when we could directly compare discount rate changes with our series for bund yield changes. The dates of these

Table 8.14 Total percentages response of US long bond yields surrounding policy turning points

	28/1/85 Tightening	11/12/85 Easing	8/4/87 Tightening	23/10/87 Easing
Change in Fed funds (%)	8.25 to 8.5	8.06 to 7.875	6.125 to 6.375	7.5 to 7.06
Response ^a (%)	8	151.4	176	86.4

	29/3/88 Tightening	6/6/88 Easing	4/2/94 Tightening	Mean response
Change in Fed funds (%)	6.56 to 6.75	9.75 to 9.5	3.0 to 3.25	91
Response ^a (%)	36.0	72.0	108.0	

^a Based on total response of benchmark 10-year bond yield two days either side of the turning point as a proportion of the policy change.

Table 8.15

	27/08/82	29/06/84	16/08/85	01/07/88	15/09/92
Discount rate change	-0.50	+0.50	-0.50	+0.50	-0.50
Bund yield change	-0.01	0.00	-0.16	+0.12	-0.13

five cases of reversals (each of 0.5 per cent) and the associated bund yield change in basis points, are shown in Table 8.15.

The average absolute size of change of bund yields in these five cases is 0.085 per cent, which compares with a weekly average, for absolute changes in bund yields, of 0.08 also (and a daily average of 0.03). There is, therefore, nothing in these data to support the hypothesis that market anticipation makes German bund yields adjust particularly sharply in response to changes that reverse the direction of discount rates.

One possibility that we considered was that discount rate reversals may have been lagging indicators of the policy shift. This shift might have been already signalled by prior changes in repo rates. We sought to test this alternative hypothesis by looking at all cases where the OMT rate change involved a reversal of direction, while measuring the accompanying bund yield change over the same windows. The results were no more encouraging. In the thirty instances of direction reversal in OMT rates during this period, only seventeen were accompanied by changes in bund yields in the same direction. In those seventeen cases, where the accompanying bund yield change was in the 'correct' direction, the absolute average size of change in bund yields over the windows (first weekly; then two days before and after) was 0.06, which again compares with an average absolute weekly (daily) change over the whole period of 0.08 (0.03).

The evidence, on the basis of this rough and preliminary test, is that there is no obvious sign of jumps in the German bund market on the occasion of reversals of direction in official short-term rates. In view of the evidence that bond yield jumps do appear to occur on such occasions in the USA and the UK, and the extent to which Bundesbank rate smoothing should lead to predictability by the market, this is a somewhat surprising negative result, particularly so in the light of the willingness, already noted in section 8.1, of some German experts to accept this general story. But this exercise was rough and ready; more work on this matter could usefully be done before we can really conclude that the bund market appears unable to predict Bundesbank smoothing operations.

8.3 CAN WE IMPROVE THE STRUCTURE OF THE SYSTEM?

What is a serious problem for an exercise of this kind is that the exact rationale for smoothing, wishing to limit the size of interest rate jumps in

any period, has never been clearly spelt out. As demonstrated in the Appendix, it must be owing to having some additional subsidiary objective in the authorities' objective function, besides price stabilization. Our evidence suggests that that is just as (or more) true of Germany as of any of our other four countries. This other objective could be output stabilization, or simply a dislike of interest rate reversals for their own sake. As one commentator, Professor Kashyap, at the Kronberg Conference said, such reversals could make the central banker look indecisive and uncertain of direction (whereas actually the underlying reality could be the exact opposite).

A related possibility is that the problem is that monetary policy actually affects ultimate target variables, e.g. nominal incomes, with a long lag, but that the monetary authorities, especially the politicians, believe that interest rate increases can only be justified by clear *present* evidence of significant deviations of actual from target outcomes. Given the inflationary bias which such political reluctance to tighten will entail, the central bank in turn may refuse to lower interest rates until there is clear present evidence of the target variable falling below its desired level. So the need to justify interest rate changes by current, rather than forecast, deviations of the target variable from its desired value may be responsible for (some part of) the observed persistence of small interest rate movements of similar sign.

If that is the case, and the results in terms of deviations of inflation from its desired level are thereby worsened, how can we best improve and repair the structure of the system? There are, perhaps, three main approaches that may be followed (they are *not* mutually exclusive). They are education, independence and proper accountability.

The aim of education would be to explain why, because of the long lags involved in the operation of monetary policy, policy actions now need to be conditioned on forecasts of the future, rather than clear observations of current deviations. This will need to be so even though forecasts are known to be highly fallible. By the same token, frequent reversals in the direction of interest rate changes, which should, perhaps, be bigger on average than in the past, possibly connected with revisions to future forecasts, should not be regarded as a sign of the authorities' failure, but as a sign of their success. This may be quite difficult to achieve. If, for example, interest rates followed the time path, $-\frac{1}{2}$, $+1$, $-\frac{1}{2}$, there would be an obvious temptation for journalists to claim that the intervening rise of 1 per cent was 'unnecessary'. Somewhat similarly, there are some private agents in financial markets who may benefit from the authorities' persistence, since their better understanding of this trait may bring them excess returns. If the authorities can act more expeditiously to keep future inflation (or monetary aggregates, or whatever target) in line with their objective, interest rate changes should depend more on the authorities' perceptions of current shocks (and less on playing 'catch-up'); hence it would be desirable if the time path of short-term interest rates should become more akin to a random walk. But that would make such changes less predictable.

It would also make interest rate changes less easy to justify. Insofar as one was appealing to the predicted effect of a current shock, e.g. to the exchange rate or unit labour costs, on future inflation, rather than to an observed current actual rise in inflation, the authorities' decision may be subject to attack under several headings: that their model may be incorrect or too uncertain to justify a policy change, that the shock is temporary and may be reversed in future, so it would be better to wait, etc.

The greater difficulty of justifying policy measures based on a forecast, rather than a currently observed deviation from target, provides yet another reason for delegating responsibility for the implementation of such policy (instrument independence in Fischer's terminology) to an independent central bank. With such an independent central bank being cordoned, to a greater degree, from the political process, it should be able to condition interest rate changes optimally on future expected economic developments, not on current observed outcomes.

If that was so, however, one might have expected the time path of German interest rates to be rather different from the more politically constrained central banks in the USA, the UK and Japan; nearer to random walk and possibly, with larger step changes. In practice the reverse is true. I speculated on some of the possible reasons for this at the end of section 8.2, but I am far from sure that we have a clear understanding of this.

Even an independent central bank, however, has to be politically sensitive. What has been established by one treaty or Act of Parliament can be reversed by its successor. The desire to maintain an existing independence against potential legislative threat can make a nominally 'independent' central bank almost as politically subservient as one without any such independence. Moreover, central bankers are, in my experience, as sensitive to public criticism as anyone else. So the pressures to defer interest rate increases, and to minimize them once taken, the 'too little, too late' syndrome, will even act on independent central banks.

That, once again, provides a reason for making the goals of monetary policy, e.g. inflation at some point in the future, both transparent and quantified, so that we have some measure of the central bank's success, or failure.¹⁵ That quantification by itself would raise the central bank's accountability and by the same token its determination to succeed. If the governors' remuneration or reappointment were furthermore directly linked to such measured success, or failure, as has now been advocated by academics (Persson and Tabellini, 1993; Walsh, 1995) as an antidote to time inconsistency, it would further strengthen central bankers' backbones.

APPENDIX: A THEORETICAL ANALYSIS OF INTEREST RATE SMOOTHING

It is difficult to discover what combination of objective function, for the monetary authority, and model of the economy could lead to the practice of interest rate smoothing, as apparently practised in reality by central banks. Let us study a few models.

Model 1: the central bank has a single objective

Let us start with a simple model, in which the target variable, π , which the central bank wishes to control follows a slightly dampened autoregressive time path, (perhaps around a deterministic trend), with shocks occurring to that path. Second, we shall assume that the monetary instrument, I , affects π with a lag.

If we start with the simplest possible first-order, single-lag system, we would have:

$$\pi_t = \alpha + b\pi_{t-1} + \gamma I_{t-1} + u_t \quad (1)$$

where b is (slightly) less than unity, and $u = (0, \sigma_u^2)$. We shall assume that the equilibrium value of π is also the desired value (no problems of time inconsistency), and that the central bank can observe u_t , but cannot offset its instantaneous effect.

Let us also start with a simple objective for the central bank,

$$U_t = -(\pi_t - \pi^*)^2 \quad (2)$$

That is, the central bank will set its I_t so as to minimize the deviation of π_{t+1} around π^* , where π^* is the equilibrium value of π .

Since π_t , defined in equation (1), exhibits persistence, maximizing U_t in a multi-period framework looks like a dynamic programming problem represented as follows:

$$\begin{aligned} \max_{I_{t-1}} E \sum_{t=1}^{\infty} \beta^t U_t \\ \text{subject to (1)} \end{aligned}$$

where β is the authorities' discount factor. This dynamic programming problem has one state variable, π_t , and one control variable, I_t . The solution can be obtained by solving the following problem:

$$V(\pi_{t-1}) = E \max_{I_{t-1}} \{ -(\pi_t - \pi^*)^2 + \beta V(\pi_t) \}, \quad (3)$$

subject to (1). The first order condition of the discretionary solution is:

$$\begin{aligned} -2\gamma(\pi_t - \pi^*) + \gamma\beta V'(\pi_t) &= 0, \\ (\pi_t - \pi^*) - 0.5\beta V'(\pi_t) &= 0, \end{aligned} \quad (4)$$

where $V'(\pi_t)$ is the first order derivative of $V(\pi_t)$ respective to π_t .

For a linear-quadratic problem such as this, $V(\pi_t)$ must also be quadratic. Without losing generality, we can write $V(\pi_t) = a_0 + a_1\pi_t + 0.5a_2\pi_t^2$, so that

$$V'(\pi_t) = a_1 + a_2\pi_t. \quad (5)$$

Substituting (5) into (4) and then taking expectations, we have

$$E\pi_t = (2\pi^* - \beta a_1)/(2 + \beta a_2),$$

hence,

$$EI_{t-1} = [E\pi_t - (\alpha + b\pi_{t-1})]/\gamma = [(2\pi^* - \beta a_1)/(2 + \beta a_2) - \alpha - b\pi_{t-1}]/\gamma. \quad (6)$$

The value of a_1 and a_2 can be solved by applying the envelope theorem to (3). Doing so gives us:

$$V'(\pi_{t-1}) = a_1 + a_2\pi_{t-1} = E\{[-2(\pi_t - \pi^*) + \beta(a_1 + a_2\pi_t)]b\}.$$

Using (4), the RHS of the above equation is obviously zero. Therefore, $a_1 = a_2 = 0$, and as a result, $E\pi_t = \pi^*$ and $EI_{t-1} = (\pi^* - \alpha - b\pi_{t-1})/\gamma$. Although the above exercise looks like a dynamic programming problem, it really boils down to a one-period problem.

Since u_t is a purely stochastic variable, unpredictable in advance, and I_t affects π with a one period lag, the time path of I_t should follow a similar stochastic path, with the pattern of reversals and continuations of the sign of I_t being similar to u_t , only with a one period lag. This is our first finding, that in a simple model, but one with persistence and lags and with the central bank *caring only about price stability*, the optimal time path of interest rates should mirror that of the inflationary shocks hitting the economy. If that is i.i.d., then so should be the time path of changes in interest rates. With the central bank caring only about price stability, so that the central bank has a single objective, theoretically there should be no interest rate smoothness outcome. Starting from this point, we are led to pursue new models with the central bank having multi-objectives.

Model 2: the central bank has multi-objectives

We assume that the central bank now has multi-objectives; it wants to achieve output stability in addition to its initial price stability as in model 1. Its new utility function becomes:

$$U_t = -(\pi_t - \pi^*)^2 - k(y_t - y^*)^2. \quad (7)$$

With such a utility function, we need also to enrich the model to include y_t in the model. Let us assume that y_t can be described by a Lucas supply function:

$$y_t = \alpha(\pi_t - E\pi_t) + \theta y_{t-1} + u_t. \quad (8)$$

With persistence, a portfolio balance effect should be included in the demand for money. By setting interest rate I_t , rather than money supply, the monetary authorities have¹⁶:

$$\pi_t = I_t + y_t - y_{t-1}. \quad (9)$$

Substituting it into (8) and after rearrangement, we obtain:

$$y_t = \alpha'(I_t - EI_t) + \theta y_{t-1} + u_t/(1 - \alpha). \quad (10)$$

where we denote $\alpha' = \alpha/(1 - \alpha)$ for simplicity. From this equation, we have

$$\pi_t = (I_t - \alpha EI_t)/(1 + \alpha) + (1 - \theta)y_{t-1} + u_t/(1 - \alpha). \quad (11)$$

We know that the central bank's problem is:

$$\max_{I_t} E \sum_{t=1}^{\infty} \beta^t U_t$$

subject to (10)

This is a dynamic programming problem with one state-variable, y_{t-1} , and one control variable, I_t . The solution can be obtained by solving the following problem:

$$V(y_{t-1}) = E \max_{I_t} \{ -(I_t + y_t - y_{t-1} - \pi^*)^2 - k(y_t - y^*)^2 + \beta V(y_t) \}, \quad (12)$$

subject to (10). The first order condition of the discretionary solution is:

$$-2(1 + \alpha')(I_t + y_t - y_{t-1} - \pi^*) - 2k\alpha'(y_t - y^*) + \alpha'\beta V'(y_t) = 0,$$

or

$$(I_t + y_t - y_{t-1} - \pi^*) + k\alpha(y_t - y^*) - 0.5\alpha\beta V'(y_t) = 0, \quad (13)$$

where $V'(y_t)$ is the first order derivative of $V(y_t)$ respective to y_t .

For a linear-quadratic problem such as this, $V(y_t)$ must also be quadratic. Without losing generality, we can write $V(y_t) = a_0 + a_1 y_t + 0.5 a_2 y_t^2$, so that $V'(y_t) = a_1 + a_2 y_t$.

Substituting $V'(y_t) = a_1 + a_2 y_t$ and (10) into (13) and then taking expectations, we have

$$EI_t = \pi^* + \alpha k y^* + 0.5 \alpha \beta a_1 - [1 + (\alpha k - 0.5 \alpha \beta a_2 - 1)\theta] y_{t-1}.$$

This can be rewritten briefly as:

$$EI_t = \pi^* + b + c y_{t-1}, \quad (14)$$

Hence,

$$I_t = EI_t + du_t = \pi^* + b + c y_{t-1} + du_t, \quad (15)$$

$$b = \alpha k y^* + 0.5 \alpha \beta a_1, \quad (16)$$

$$c = - [1 + (\alpha k - 0.5 \alpha \beta a_2 - 1)\theta], \quad (17)$$

$$d = - (\alpha k - 0.5 \alpha \beta a_2 - 1)/(\alpha^2 k - 0.5 \alpha^2 \beta a_2 + 1). \quad (18)$$

Applying the envelope theorem gives us:

$$V'(y_{t-1}) = a_1 + a_2 y_{t-1} = E \{ [-2k(y_t - y^*) + 2(i_t - y_t + y_{t-1} - \pi^*) + \beta(a_1 + a_2 y_t)](\theta - \alpha'c) - 2(i_t - y_t + y_{t-1} - \pi^*) \}.$$

Substituting y_t and regrouping gives us:

$$a_2 = \theta(\theta - \alpha'c)(-2k + \beta a_2) + 2(\theta - \alpha'c - 1)(c - \theta + 1), \quad (19)$$

$$a_1 = [(\theta - \alpha'c)(-2k + \beta a_2 - 2) + 2](1 - \theta)y_n + (\theta - \alpha'c)(2ky^* + \beta a_1 + 2b) - 2b. \quad (20)$$

a_2 and a_1 can be solved by substituting b and c into (19) and (20). The solution is quite complicated. A simpler solution is to replace a_2 in (19) by its expression in c , then solve for c . Obviously, after obtaining c , we can substitute it into (17) and (18) respectively to find out a_2 and a_1 .

The critical implication of this, compared with model 1 where the monetary authorities have only a single objective, is that when the monetary authorities have multi-objectives, their interest rate instrument, I_t , does not follow a random walk but rather moves smoothly. This can be seen clearly from the optimal feedback rule, equation (15), where I_t is a function of y_{t-1} which moves smoothly owing to its persistence. Intuitively, as persistence allows above average employment to continue, the authorities will place more weight on holding inflation down, and vice versa, and all this can only be achieved by adjusting the instrument I_t smoothly rather than letting it to follow a specific pattern of random walk as in model 1.

Let us assume that $y_t = \bar{y}_y < y_n$ is a starting point. We can then work out both the current and expected future trajectory of both $I_t \rightarrow I_{t+m}$ and $y_t \rightarrow y_{t+m}$, with $E u_{t+m} = 0$. By substituting $E I_t$ into the Lucas equation (15), we have

$$y_t = \theta y_{t-1} + [(\alpha d + 1)/(1 + \alpha)] u_t. \quad (21)$$

Therefore,

$$y_{t+m} = \theta^m \bar{y}_t + \frac{\alpha d + 1}{1 + \alpha} \sum_{j=0}^{m-1} \theta^j u_{t+m-j} = \theta^m \bar{y}_t + \frac{\alpha d + 1}{1 + \alpha} \sum_{j=0}^{m-1} \theta^j u_{t+m-j}$$

and $E y_{t+m} = \theta^m \bar{y}_t$. It is clear that the speed of convergence for y_t is determined by the value of θ .

From (15), $I_t = \pi^* + b + c y_{t-1} + du_t$. Similarly, I_t also converges to its

equilibrium smoothly and gradually but its speed of convergence is determined by the value of θ and c , the latter depending on many parameters, including α , β , k and θ . It can also be shown that π_t also moves smoothly and its speed of convergence is determined by the value of θ and c .

Another way to model the monetary authorities having multi-objectives is simply to include the interest rate adjustment into their utility function. That is setting:

$$U_t = -(\pi_t - \pi^*)^2 - k(I_t - I^*)^2. \quad (22)$$

Either with the simplest possible first-order, single lag system, $\pi_t = \alpha + b\pi_{t-1} + \gamma I_{t-1} + u_t$, as defined in (1) in model 1, or with a more complicated system as defined by (8) and (9) in model 2, the outcome would have the property of interest rate smoothing. Of course, the 'smoothness' of interest rate smoothing varies, depending on the parameter configuration of the model.

What is a serious problem for this exercise is that the exact rationale for wishing to limit the size of interest rate jumps in any period has never been clearly spelt out. The outcome of interest rate smoothing could be caused by output stabilization, or simply interest rate instrument stabilization, in addition to the 'primary' objective of price stabilization. Yet it does seem to be the case that some additional subsidiary objective *does* enter the authorities' objective function.

NOTES

- 1 McCallum (1994a, b) has also emphasized that relationships between spot and forward rates, in both the foreign exchange and money markets, will be affected by the interaction between the authorities' policy actions and reactions and those of private sector agents. This can make it extremely difficult to make sense of (econometric) studies of such markets without a careful and thorough identification of the authorities' actions and objectives.
- 2 Since the Fed has a short-term interest rate target but allows daily deviations to occur, the very short-term yield curve, overnight and three-month, has predictive ability. Again, since the Fed smooths over periods of a month, or so, the one-month and two-month yield gap has predictive ability. But, since its short-term smoothing behaviour is incorporated into three-month market rates, and beyond that horizon the Fed reverts to random-walk behaviour, there is no predictive ability in the three-month and six-month yield gap. See Rudebusch (1995, pp. 265–71).
- 3 The sample was chosen to include large countries which did not (for most of the period) peg their exchange rates against another country's, i.e. they were able to exercise an independent monetary policy.

Thus the similar behaviour was not caused by each of these other countries simply following another, e.g. the USA, as could happen in a fixed exchange rate regime.

The contemporaneous correlations between interest rates (money market

rates, as described on p. 124, in levels 1983 M1 to 1993 M12, except for correlations involving Australia, when the data period starts in 1984 M1) were as follows.

	Australia	Germany	Japan	UK	USA
Australia					
Germany	−0.475				
Japan	0.335	0.423			
UK	0.711	0.200	0.592		
USA	0.709	−0.089	0.472	0.515	

In view of the extent of common global factors (e.g. oil shocks), we regard these values as quite low, indicating a considerable degree of national independence in setting monetary policy.

When we regressed the *changes* in each country's money market interest rates (same data period) against the prior three lagged values of its own changes and the prior three lagged values of changes in each of the other country's money market rates, the only significant influences of other countries' prior rate changes on own country interest rate changes were as follows:

- (a) Bilateral causality between Germany and the UK, with the prior month's change in the other country's money market rate having a significant (positive) impact on the current change in the domestic own rate.
 - (b) Some weak bilateral causality between the UK and Australia whereby prior changes in Australia's money market rates, lagged two and three months, marginally affect the UK, and vice versa.
 - (c) Some weak evidence of one-way causation from prior changes in Japanese money market rates to both UK and Australian money markets.
 - (d) Some very weak indication of causality from prior German and UK money market rate changes to US money market rates.
- 4 When the US inflationary experience deteriorated at the end of the 1970s, the Fed adopted a new monetary regime, of targeting unborrowed reserves, that forced their interest rate adjustments to become more aggressive. Rudebusch excludes this period from his data set, since the Fed funds rate was not then the prime intermediate target of monetary policy. Nevertheless, this reinforces the view that, in a larger sense, the authorities' policy is endogenous to circumstances and should not be viewed as exogenously determined.
 - 5 We would like to thank the European Office of the Australian Reserve Bank, the Deutsche Bundesbank, the Bank of Japan, the Bank of England and the Bank for International Settlements for their assistance with the provision of data.
 - 6 Durations between rate changes measured in working days, not counting weekends. For the sake of simplicity of computation, and because of incomparabilities of holidays across countries, they were disregarded with two exceptions: 24–26 December and 1 January are assumed to be holidays.
 - 7 We sought to test whether the central locations of both the sizes of and the duration between these discount rate changes, were significantly different from each other (i.e. whether we could accept the null of similarity) using the Mann–Whitney non-parametric test (based on relative rankings). This is preferable to the Wilcoxon test when one is not comparing matched pairs (see Newbold, 1991). We did not undertake similar tests for the money

market rate series, nor for any series based on monthly averages.

The results were, (Y, accept similarity at 5 per cent significance; N, reject; n.a., too few observations for the test, or not appropriate; S, test for size of change; D, test for duration between changes; C, continuations; R, reversals), as follows:

	1 + + = - -		2 + - = - +		3 C = R	
	S	D	S	D	S	D
UK base rates	N	Y	N	Y	n.a.	N
US discount rates	Y	Y	Y	Y	Y	N
German discount rate	N	Y	n.a.	n.a.	n.a.	n.a.
Australian discount rate	Y	Y	Y	Y	N	N

- 8 When more than one rate change took place during the course of a day, as in the UK on 16 September 1992, we always took the end-daily figure.
- 9 I am grateful to Mr A. Haldane for reminding me of this point at the Kronberg Conference.
- 10 The choice of the appropriate series for inflation *ought* to be much more controversial than it usually is. For example, when using the CPI or RPI as the measure of inflation, Japan's record in controlling inflation over the last decade is extremely good (see Table 8.8). Yet this was the period of asset price boom and bust. Few outside commentators would describe Japan's recent monetary policies as exemplary; yet by the statistical standards that are normally applied, as will be done here, they will be counted as such!
- 11 In practice, the precise numerical value chosen for π^* makes little difference in most of the subsequent regression exercises, since it only affects the constant term, not the other coefficients.
- 12 Charts for the relationships between money market rates and inflation are available on request. Since they show qualitatively the same relationship, they are excluded in order to save space.
- 13 Note that this possible explanation for interest rate smoothness is diametrically opposite to that suggested by Cukierman (1992, Chapter 7), which is that the authorities are trying to avoid disturbances to the financial system. Insofar as smoothing short rates leads to a particularly abrupt change in long rates, on the occasion of reversals, it would then lead to abnormally large, and sudden, valuation changes at such times. The events in bond markets following the US Fed's reversals in February 1994 could be interpreted in this light.
- 14 Since all the reversals involved changes of 0.5 per cent, and this was also nearly the average size of continuations during this period, we did not find it necessary also to adjust for the size of change.
- 15 Two issues, not immediately relevant to the main text, should be briefly raised here. First, such goals *should* be set by the democratically elected government, not by the central bank itself. The central bank should not have 'goal independence', to use Fischer's (1994) terminology.

Second, it is often argued that a final objective for inflation is inappropriate because the central bank cannot control it. Some supply shocks (e.g. indirect taxes, food and energy prices) may be handled by an appropriate choice of price index. But inflation will always be affected by current shocks, which, owing to lags, the authorities cannot offset. So what? Businesses are similarly affected by exogenous shocks, and no one uses that as an argument why their executives should not be assessed and paid in terms of their current profits. The problem of short-term control should

properly be used in connection with choosing the appropriate band width around the target, but is not crucial in the choice of what is the best target.

But, again, I have discussed such issues at some length in various papers on central bank independence (e.g. Goodhart and Viñals, 1995), and I would not want to highlight them for discussion here.

- 16 As earlier noted, we do not intend to discuss money supply as the control instrument, or the differences between these two instruments for our simple model here.

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9

COMMENT ON CHAPTER 8

Otmar Issing

Central bankers are often accused of seeing inflationary dangers around every corner and searching out grounds to raise interest rates. It is therefore almost a relief to be asked, by such a distinguished monetary policy expert (and former central banker) as Mr Goodhart, to 'plead guilty to the charge of "too little"'. The case is presented in the chapter using a wide selection of evidence and imaginative techniques. But my intention is not to act for either the prosecution or the defence, but rather to serve as *amicus curiae* by elucidating some aspects of the German financial system that may help to explain the results in Chapter 8. In particular, I shall concentrate on perhaps the most surprising result, namely that although the central banks of Germany and Japan have had the best records in keeping inflation low and stable, they also seem to be the most prone to changing official nominal interest rates in a series of small steps ('too little'), and only once any trend in the inflation rate has begun to be reversed ('too late').

One question to be asked is whether the same story emerges when one looks at changes in other interest rates, and at other time horizons, than those considered in the paper. For example, the Deutsche Bundesbank implements monetary policy with three official interest rates: rates at the Lombard and discount facilities and the rate governing repurchase operations (*Pensionsgeschäfte*). The Lombard rate provides a ceiling on short-term money market rates, and the repurchase rate effectively guides money market rates over the short term. The Lombard and discount rates can be changed at most every two weeks following the regular meetings of the Bundesbank Council, but are in practice changed only infrequently when the Bundesbank wishes to create room for major changes in the level of interest rates. The repurchase rate set in weekly tenders is usually adjusted gradually along the trend of interest rate movements.¹ On the one hand, these changes exploit the room created by a shift in the band formed by the Lombard and discount rates. On the other hand, when the repurchase rate attains a certain minimum distance from the upper or